Spatial–Spectral Diffusion Contrastive Representation Network for Hyperspectral Image Classification

Yimin Zhu and Linlin Xu, *Member, IEEE*

***Abstract*— Although efficient extraction of discriminative spatial–spectral features is critical for hyperspectral image classification (HSIC), it is difficult to achieve these features due to factors such as the spatial–spectral heterogeneity and noise effect. This article presents a spatial–spectral diffusion contrastive representation network (DiffCRN) based on the denoising diffusion probabilistic model (DDPM) combined with contrastive learning (CL) for HSIC, with the following characteristics. First, to improve spatial–spectral feature representation, instead of adopting the UNet-like structure, which is widely used for DDPM, we design a novel staged architecture with a spatial self-attention denoising (SSAD) module and a spectral group self-attention denoising (SGSAD) module in DiffCRN with improved efficiency for spectral–spatial feature learning. Second, to improve unsupervised feature learning efficiency, we design a new DDPM model with logarithmic absolute error (LAE) loss and CL that improve the loss function effectiveness and increase the instance-level and interclass discriminability. Third, to improve feature selection, we design a learnable approach based on pixel-level spectral angle mapping (SAM) for the selection of timesteps in the proposed DDPM model in an adaptive and automatic manner. Last, to improve feature integration and classification, we design an adaptive weighted addition module (AWAM) and a cross-timestep spectral–spatial fusion module (CTSSFM) to fuse timestepwise features and perform classification. Experiments conducted on widely used four hyperspectral image (HSI) datasets demonstrate the improved performance of the proposed DiffCRN over the classical backbone models and the state-of-the-art generative adversarial network (GAN), transformer models, and other pretrained methods. The source code and pretrained model will be made available publicly.**

***Index Terms*— Adaptive selection of diffusion timestep, denoising diffusion probabilistic model (DDPM), feature extraction, hyperspectral image classification (HSIC), unsupervised feature learning.**

NOMENCLATURE

Notations	*Descriptions*
H, W, and C	Height, width, and number of spectral channels, respectively.
$\mathcal{X} \in \mathbb{R}^{H\times W\times C}$	Hyperspectral image cube.
$\mathcal{H}_0 \in \mathbb{R}^{P\times P\times C}$	Cropped original spectral–spatial instance using a $P \times P$ neighborhood.
T	Total number of Gaussian diffusion steps.
$\mathcal{H}_t \in \mathbb{R}^{P\times P\times C}$, $t \in \{1, \ldots, T\}$	Latent feature variable at the tth timestep generated by diffusion forward process; specifically, $\mathcal{H}_T$ is nearly an isotropic Gaussian distribution.
ϵ_θ	Spatial–spectral denoising network with parameters θ.
$\mathcal{F}$	Feature selection function
$\mathscr{A}_\alpha$	Contrastive learning module $\mathscr{A}$ with parameters α.
$\mathscr{C}_\beta$	Classification network $\mathscr{C}$ with parameters β.
$y_i \in \{1, \ldots, N\}$, $i \in \{1, \ldots, H \times W\}$	Class label of each pixel, where N represents the number of category.

Received 24 January 2025; revised 3 June 2025; accepted 2 July 2025. Date of publication 7 July 2025; date of current version 25 July 2025. This work was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) under Grant RGPIN-2019-06744. *(Corresponding author: Linlin Xu.)*

The authors are with the Department of Geomatics Engineering, University of Calgary, Calgary, AB T2N 1N4, Canada (e-mail: yimin.zhu@ucalgary.ca; lincoln.xu@ucalgary.ca).

Digital Object Identifier 10.1109/TGRS.2025.3586287

I. INTRODUCTION

HYPERSPECTRAL image classification (HSIC) as a fundamental task in hyperspectral image (HSI) processing aims to classify each pixel's in HSI to achieve semantic classification maps, which is critical to support various applications, e.g., land cover and crop mapping [1], [2], environmental monitoring [3], [4], and mineral detection and mapping [5], [6]. Although HSIC is critical to support various environmental applications, achieving efficient HSIC is a challenging task. Spectral–spatial heterogeneity reflected by the tradeoff on spectral and spatial resolutions, low signal-to-noise ratio, mixed pixels, and the lighting effect [7] requires enhanced methods to extract discriminative spatial–spectral features from subtle class signatures of different land cover and land use classes [8]. Therefore, many machine learning models have been proposed to improve the modeling of the complex spectral–spatial correlation and heterogeneity in HSI with unsupervised [9], semisupervised [10], and supervised approaches [11].

Comparing with traditional machine learning methods, the deep learning (DL) methods have demonstrated improved spatial–spectral feature learning capability based on deep convolutional neural networks (CNNs) [12], [13], [14], [15], [16]. Besides, other networks are exploited constantly to overcome the limitation of CNNs, such as transformers [17], [18], [19], [20], graph neural networks [21], [22], capsule networks [23],

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

[24], and multiview learning [25], [26]. However, most of these approaches are supervised methods, which rely predominantly on training samples to learn the explicit mappings between semantic labels and spatial–spectral data, leading to drawbacks such as high labeling and data preparation cost, as well as poor generalization capability outside the discriminant space. As a result, some recent works [27], [28], [29] tend to exploit semisupervised learning combining manifold learning, active learning, and the pseudolabel method for HSIC with optimizing abundant unlabeled data to characterize spectral–spatial relations.

Unsupervised feature extraction is another feasible approach to model the spatial–spectral correlation in HSI for HSIC. Autoencoder (AE) is a common unsupervised framework, which provides a probability to an input, and is designed to extract features hierarchically by minimize the reconstruction error, which has been widely used for HSIC [30], [31], [32], [33], [34]. However, the network layers are completely connected, which contributes to high computing resources. Some enhanced AE approaches are applied to HSIC, e.g., fully conv–deconv network with residual learning [35] and the hypergraph-structured AE [36]. Considering that different classes may have different spectral shifts, domain adaptation (DA) is used to solve the problem of distribution alignment to further construct the spectral–spatial relationships for enhanced unsupervised HSIC [37]. Furthermore, the generative adversarial network (GAN) is introduced to train a DL-based feature extractor in an unsupervised manner [38], [39], [40]. Although the above unsupervised feature learning frameworks can excavate implicit spectral–spatial relationships, the GAN-based framework may lead to model collapse due to the complex training procedure [41], [42].

Recently, the denoising diffusion probabilistic models (DDPMs) [43], as unsupervised feature learning approaches, have garnered significant research interest owing to their notable advantages over alternative approaches, including GAN-based methods [44]. As generative models, DDPMs train a denoising AE to learn the reverse of a Markovian diffusion process [45]. Recently, DDPMs have demonstrated superiorities in vision tasks such as image generation [43], [46], [47], [48], image super-resolution [46], [49], [50], segmentation [51], [52], [53], [54], [55], [56], and classification [57], [58], [59], [60].

DDPMs have a strong potential to improve HSIC by providing enhanced feature extraction from HSI due to the following reasons. First, DDPMs provide a principled probabilistic framework for feature learning, which can explicitly model the uncertainty and variability in HSI, especially at the reverse process of the diffusion model, therefore enabling the extraction of improved spectral–spatial features for HSIC. Unlike the traditional denosing model, e.g., total variation (TV) model [61], a predefined model, means deterministic models and not data-driven, leading to no uncertainty estimates. Some CNNs and wavelet-based methods [62] provide point estimates without uncertainty measures. Second, DDPMs can capture both high- and low-level features that can enhance HSIC using proper timesteps in DDPMs [51]. Third, DDPMs have a more stable training mechanism and are less prone to model collapse compared to other deep unsupervised models, such as the GAN-based model [51]. Given the theoretical advantages of DDPMs, they have not been sufficiently studied for enhanced HSIC. Although some researchers have used DDPMs for HSIC [54], [59], [60], there is still a lack of research on spatial–spectral DDPMs that are tailor-designed to HSI characteristics and challenges. The existing DDPM-based methods for HSIC have the following deficiencies.

1) *Feature Redundancy:* When viewing DDPM as feature extraction, the feature under the index of diffusion timestep and the model layer is used for enhancing the spatial–spectral information, as the case in [59] and [60]. In [59], spectraldiff uses manually selected features at specific diffusion timesteps and Unet layers by performing many ablation experiments to prove the effectiveness of diffusion features, and the method in [63] selects feature from predefined timestep list [5, 10, 50, 100, 200, 400], which may not be practical, even though these methods give us general directions to select features. In [60], the authors use all of the timesteps and design a specific and complex selective timestep network to refine the feature, which also leads to data redundancy. The above two methods do not propose an efficient, adaptive, and simple way to discard and select the timesteps that are most informative.
2) *Classification Boundary Smoothing:* Chen et al. [59], Zhou et al. [60], and Zhang et al. [63] use Unet as the backbone to extract spatial–spectral features, but the inductive bias within CNNs will lead to rigid constraints on kernel weight, forming smooth boundaries, and blurred high-frequency information. From the perspective of land cover mapping, clear category boundaries are essential. Thus, it is necessary to explore new network architecture, i.e., transformer, for DDPM-based feature extraction.
3) *Insufficient Utilization of Theoretical Advantages of DDPM:* As the newest and promising generative model, DDPMs have a lot of good mathematical properties, i.e, reconstruction. While few methods fully take advantage of DDPMs to perform contrastive learning (CL) between two views, i.e., reconstructed patch and raw patch, to enhance the diversity and richness of spectral–spatial feature information.
4) *Loss Function Design:* Even though the design of the loss function is very concise and derives from rigorous mathematical derivations, i.e., evidence lower bound (ELBO), the variance of added noise on different timesteps varies from each other. The variance level increases gradually with the passage of the diffusion timestep. The related works in [51], [59], [60], [63], [64], and [65] use the mean squared error (mse) function to update the model parameter, which equally treats the noise under different sets of timesteps. Thus, considering the variance schedule, a nonuniform loss function should be designed for the diffusion model.

This article, therefore, presents a new spatial–spectral diffusion contrastive representation network (DiffCRN) based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

on DDPM combined with CL for HSIC, with the following contributions.

1) To improve spatial–spectral feature representation in DiffCRN, instead of adopting the UNet-like structure, which is widely used for DDPMs, we design a novel staged architecture with a spatial self-attention denoising (SSAD) module and a spectral group self-attention denoising module (SGSAD) in DiffCRN with improved efficiency for spectral–spatial feature learning.
2) To improve unsupervised feature learning efficiency, we design a new spatial–spectral DiffCRN with logarithmic absolute error (LAE) loss and CL that improve the loss function effectiveness and increase the instance-level and interclass discriminability.
3) To improve feature selection, we design a learnable approach based on pixel-level spectral angle mapping (SAM) for the selection of timesteps in the proposed spatial–spectral DiffCRN in an adaptive and automatic manner.
4) To improve feature integration and classification, we design an adaptive weighted addition module (AWAM) and cross-timestep spectral–spatial fusion module (CTSSFM) to fuse timestepwise features and perform classification.

Experiments conducted on widely used four HSI datasets demonstrate the improved performance of the proposed DiffCRN over the classical backbone models and state-of-the-art GAN, transformer models, and other pretrained methods. The remainder of this article is organized as follows. Section II describes related work. In Section III, our proposed DiffCRN is introduced in detail. Section IV discusses our experiment results. Some conclusions are drawn in Section V.

II. Related Work

In this section, we describe existing research directions relevant to our research based on traditional spectral–spatial feature extraction, as well as those based on DL, along with mainstream generative tasks and the development context of DDPMs. Through a comprehensive review of related research, our aim is to establish a strong foundation for our proposed method, solidify our contribution, and apply it effectively in complex HSIC research.

A. HSI Classification Based on Supervised Learning

Traditional feature extraction approaches developed on statistical computations and linear algebra methods along with the canonical machine learning methods [66], [67], [68], which ignore the spatial relationships between each spectral pixel. Although researchers have proposed various methods that take spatial structure and texture information into consideration, e.g., the morphological profiles (MPs) [69], extended MPs (EMPs) [70], gray-level co-occurrence matrix (GLCM) [71], Markov random field (MRF) [72], and invariant attribute profiles (IAPs) [73]. However, the above methods are criticized due to that the extracted shallow features lack representativeness, and the design of the hyperparameters is knowledge-driven.

During the last decade, DL based methods have flourished in HSIC. Many progressive DL-based data-driven networks have been widely utilized for supervised HSIC methods in an end-to-end manner, providing automatic feature learning from data [30], [74], [75], [76], e.g., deep belief networks (DBNs) [30]. Particularly, HSIC benefits from CNN-based models due to the ability to extract locally spatial contextual information and spectral variability information [13], [77], [78]. Specifically, since the 3-D convolution (3DCONV) can simultaneously investigate spectral–spatial relationships [13], [15]. For instance, Zhong et al. [16] presented a spectral–spatial residual network (SSRN), which investigates the 3-D spatial–spectral features by skip-connections to ensure the generalization ability. Furthermore, Tang et al. [79] factorized the mixed feature maps by their low- and high-frequency information using a 3-D Octave convolution model to realize the interaction of spectral–spatial information. However, these 3DCONV methods contain huge model parameters, which require a large number of training samples [80]. Furthermore, many researchers have adopted transformer-based sequential approaches [19], [20], [81] to design networks that address CNNs' limitations in extracting and representing spectral signature sequence attributes. For instance, Sun et al. [82] constructed a spatial–spectral feature tokenization transformer (SSFTT) to capture spatial–spectral features and high-level semantic features. However, these works lack consideration of significant differences, such as the size and the number of basic elements between natural language processing (NLP) and HSI. They still face quadratic computation complexities [83], hence limited to modeling complex relations. According to [84], their capture of long-range dependencies even hinders network optimization.

B. HSI Classification Based on Unsupervised Learning

1) AE-Based Model: Unsupervised feature extraction methods do not need labeled data and are not restricted to specific prerequisites. Typically, AEs are common unsupervised frameworks. Kemker et al. [85] proposed a stacked convolutional AE (SCAE) to identify deeper features and yielded superior performance. Masci et al. [86] and Du et al. [87] used several AEs to learn a hierarchical feature representation, resulting in more discriminative features. The fully connected operators in AE were replaced by convolutional operators so that the network can directly extract spectral–spatial joint features from cubes. Even so, they often involve complex training processes, which also need numerous labeled samples in the classification stage.

2) GAN-Based Model: As a novel unsupervised classification scheme, the GANs have also been employed for HSIC. For instance, Zhan et al. [88] developed a framework for HSIC using a 1-D GAN for HSIC (HSGAN) using discriminator features. Zhang et al. [38] designed a deconvolutional generator and a 2-D CNN discriminator to learn spectral–spatial relationships of data sets and extract spatial–spectral features, respectively. Hang et al. [39] proposed a multitask GAN (MTGAN), which consisted of a generator network for hyperspectral imagery reconstruction and classification,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

and a discriminator network to discriminate between the real and fake (generated) data. Although GANs are powerful to energize unsupervised HSI classifiers, they always encounter the failures of model collapse and nonconvergence.

C. Development of DDPMs

DDPMs [43], which are inspired by nonequilibrium thermodynamics, have emerged as the new state-of-the-art family of deep generative models and have broken the long-time dominance of GANs, which have achieved great success in the field of natural image generation, NLP, temporal data modeling, and multimodal modeling. DDPMs are a family of probabilistic generative models that operate as parameterized Markov chains, trained via variational inference to generate samples conforming to data distributions within a finite sequence. It progressively destroys data by injecting noise, then reverses the former by learning transition kernels parameterized by deep neural networks, and generates samples that resemble the original.

DDPMs are trained by optimizing the variational lower bound of the negative log-likelihood of the data, and they avoid mode collapse often encountered by GANs. Because of the slow sample generation speed of DDPM, Song et al. [89] presented denoising diffusion implicit models (DDIM), a class of non-Markovian diffusion processes, whose reverse process can be 10× to 50× faster than DDPMs. Baranchuk et al. [51] demonstrated that the pretrained conditional DDPM can serve as a feature extractor for discriminative computer vision problems. To train neural radiance fields (NeRFs) with only a small views, Yang et al. [90] found that an unconditional diffusion prior can be treated as guidance for 3-D generation and rendering. Furthermore, DDPMs have been successful on high-level semantic understanding tasks, i.e., image super resolution, natural and medical segmentation, and classification.

Although DDPMs have been widely applied to the community of natural image generation, remote sensing fake sample generation [48], synthetic aperture radar despeckling [91], remote sensing change detection [54], seldom in the field of HSIC. Hence, the critical research question is how to adapt the latent feature representations of DDPMs for discriminative HSIC tasks, with the help of the construction of raw HSI data distribution. Considering the theoretical advantages of DDPMs, some researchers have used DDPMs for HSIC [54], [59], [60]. However, there is still a lack of research on spatial–spectral DDPMs that are tailor-made to HSI characteristics and challenges. For example, efficient selection of timesteps in DDPM in an adaptive learnable manner for HSIC has not been addressed, given the critical importance of timestep selection [51], [54], [59], [60]. In [54] and [59], the timesteps are manually selected to characterize spectral–spatial features, and Zhou et al. [60] use all of the timesteps, which might lead to redundancy.

III. Methodology

A. Problem Formulation

1) Notation: To facilitate reading and reduce ambiguity, we first define the common notation in Nomenclature.

2) Spatial–Spectral Diffusion Representation Network: The spatial–spectral diffusion representation network is a latent variable model with the form [43], [92]

$$p_\theta(\boldsymbol{\mathcal{H}}_{\mathbf{0}}) := \int p_\theta(\boldsymbol{\mathcal{H}}_{\mathbf{0:T}}) d\boldsymbol{\mathcal{H}}_{\mathbf{1:T}} \tag{1}$$

$$p_\theta(\boldsymbol{\mathcal{H}}_{\mathbf{0:T}}) := p(\boldsymbol{\mathcal{H}}_{\boldsymbol{T}}) \prod_{t=1}^{T} p_\theta(\boldsymbol{\mathcal{H}}_{t-1}|\boldsymbol{\mathcal{H}}_t) \tag{2}$$

where $\boldsymbol{\mathcal{H}}_1, \ldots, \boldsymbol{\mathcal{H}}_t, \ldots, \boldsymbol{\mathcal{H}}_T$ are latents of the same dimensionality as the spectral–spatial instance sampled from a real HSI data distribution $\boldsymbol{\mathcal{H}}_0 \sim q(\boldsymbol{\mathcal{X}})$. The joint distribution $p_\theta(\boldsymbol{\mathcal{H}}_{\mathbf{0:T}})$ is called the reverse process, and it is defined as a Markov chain with learned Gaussian transitions starting at $p(\boldsymbol{\mathcal{H}}_T) = \mathcal{N}(\boldsymbol{\mathcal{H}}_T; \mathbf{0}, \boldsymbol{I})$. Thus, the spatial–spectral diffusion representation network enables to construct hyperspectral instance distribution with hundreds of spectrums by training the denoising network in the reverse process to estimate the noise added in the spatial–spectral forward diffusion process, and then, the desired image $\boldsymbol{\mathcal{H}}_0$ will be generated.

a) Spatial–spectral diffusion forward process: As mentioned above, by performing variational inference on a Markovian process with T number of timesteps, diffusion models can convert the Gaussian noise distribution $\boldsymbol{\mathcal{H}}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ into the target distribution of $\boldsymbol{\mathcal{H}}_0$.

The spatial–spectral diffusion forward process is inspired by nonequilibrium thermodynamics, which can be viewed as a Markov chain where the noise is gradually added to $\boldsymbol{\mathcal{H}}_0$ to produce latent variables $\boldsymbol{\mathcal{H}}_1$ through $\boldsymbol{\mathcal{H}}_T$ at time t with fixed variance schedule β_t. At timestep t, the noisy spectral–spatial instance $\boldsymbol{\mathcal{H}}_t$ can be represented as follows:

$$q(\boldsymbol{\mathcal{H}}_1, \ldots, \boldsymbol{\mathcal{H}}_T|\boldsymbol{\mathcal{H}}_0) := \prod_{t=1}^{T} q(\boldsymbol{\mathcal{H}}_t|\boldsymbol{\mathcal{H}}_{t-1}) \tag{3}$$

$$q(\boldsymbol{\mathcal{H}}_t|\boldsymbol{\mathcal{H}}_{t-1}) := \mathcal{N}(\boldsymbol{\mathcal{H}}_t; \sqrt{\beta_t - 1}\boldsymbol{\mathcal{H}}_{t-1}, \beta_t \boldsymbol{I}). \tag{4}$$

Through the reparameterization trick, the hyperspectral instance $\boldsymbol{\mathcal{H}}_t$ can be sampled directly at an arbitrary timestep t in the closed form, and the marginal distribution of intermediate latent variables $\boldsymbol{\mathcal{H}}_t$ given $\boldsymbol{\mathcal{H}}_0$ can be derived as

$$q(\boldsymbol{\mathcal{H}}_t|\boldsymbol{\mathcal{H}}_0) = \mathcal{N}(\boldsymbol{\mathcal{H}}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{\mathcal{H}}_0, (1 - \bar{\alpha}_t)\boldsymbol{I}) \tag{5}$$

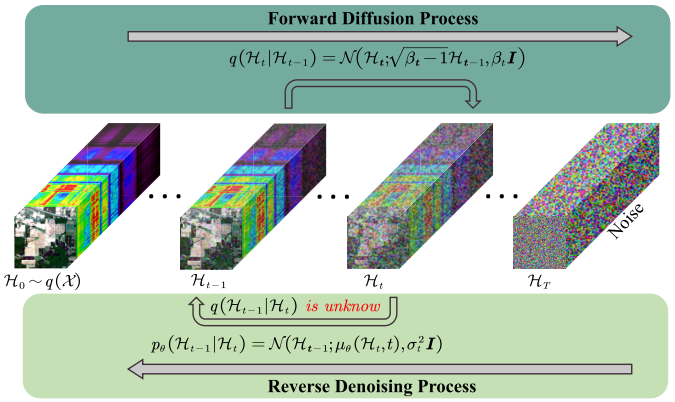
$$\boldsymbol{\mathcal{H}}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{\mathcal{H}}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \tag{6}$$

where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$, and $\epsilon \sim \mathcal{N}(0, \boldsymbol{I})$. In (4), the magnitude of the added Gaussian noise decreases as the value of α_t increases, which also means that the information of $\boldsymbol{\mathcal{H}}_0$ decreases as the timestep t increases in (6).

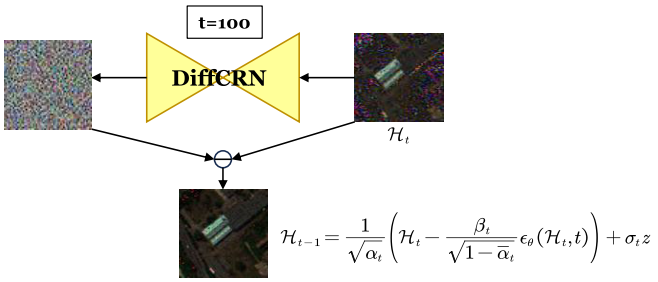
b) Spatial–spectral diffusion reverse process: The spatial–spectral diffusion reverse process, modeled by a neural network $\boldsymbol{\epsilon}_\theta$, aims to learn to remove the degradation brought from the diffusion forward process, i.e., denoising. In each timestamp t of the reverse process, the denoising operation is performed on the noisy spectral–spatial instance $\boldsymbol{\mathcal{H}}_t$ to obtain the previous instance $\boldsymbol{\mathcal{H}}_{t-1}$. In (2), since $p(\boldsymbol{\mathcal{H}}_{t-1}|\boldsymbol{\mathcal{H}}_t)$ is unknown and depends on the data distribution, we approximate it using a neural network $\boldsymbol{\epsilon}_\theta$ to predict the mean $\mu_\theta(\boldsymbol{\mathcal{H}}_t, t)$ of a Gaussian distribution as follows:

$$p_\theta(\boldsymbol{\mathcal{H}}_{t-1}|\boldsymbol{\mathcal{H}}_t) := \mathcal{N}\left(\boldsymbol{\mathcal{H}}_{t-1}; \mu_\theta(\boldsymbol{\mathcal{H}}_t, t), \sigma_t^2 \boldsymbol{I}\right) \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.



(a)



(b)

Fig. 1. (a) Denoising diffusion model forward and backward process. $q(\mathcal{H}_t|\mathcal{H}_{t-1})$, $p_\theta(\mathcal{H}_{t-1}|\mathcal{H}_t)$ represent the noise-adding forward process and the denoising backward process, respectively. The essential question is to estimate the conditional probability $q(\mathcal{H}_{t-1}|\mathcal{H}_t)$. (b) Visual illustration of (10).

where σ_t^2 is the variance of the conditional distribution $p(\mathcal{H}_{t-1}|\mathcal{H}_t)$, following:

$$\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t. \tag{8}$$

As reported by Ho et al. [43], the learning objective for the above model is derived by considering the variational lower bound, and then, the mean $\mu_\theta(\mathcal{H}_t, t)$ of the conditional distribution $p(\mathcal{H}_{t-1}|\mathcal{H}_t)$ can be reparameterized as follows:

$$\mu_\theta(\mathcal{H}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathcal{H}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathcal{H}_t, t)\right) \tag{9}$$

where $\boldsymbol{\epsilon}_\theta$ is the denoising network intended to predict added gaussion noise $\boldsymbol{\epsilon}$ from the noisy hyperspectral instance $\mathcal{H}_t$ at timestep t. The forward and reverse processes are illustrated in Fig. 1.

c) Sampling: To sample $\mathcal{H}_{t-1} \sim p_\theta(\mathcal{H}_{t-1}|\mathcal{H}_t)$ is to iteratively compute

$$\mathcal{H}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathcal{H}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathcal{H}_t, t)\right) + \sigma_t z \tag{10}$$

where $z \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, $t = T, \ldots, 1$, as shown in Fig. 1(b).

According to Tweedie's formula [93], we can use $\hat{\mathcal{H}}_0 = \mathcal{H}_{0|t} \triangleq \mathbb{E}[\mathcal{H}_0|\mathcal{H}_t]$ to represent the estimation of $\mathcal{H}_0$ at timestep t given $\mathcal{H}_t$ as follows:

$$\hat{\mathcal{H}}_0 \simeq \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathcal{H}_t + (1-\bar{\alpha}_t)\boldsymbol{\epsilon}_\theta(\mathcal{H}_t, t)). \tag{11}$$

B. Architecture and Key Components of Spatial–Spectral DiffCRN

1) Overall Architecture: Unlike UNet-like backbone architecture of the diffusion models such as [54], [59], [91], and [94], we introduce a stage architecture with a spatial–spectral dual self-attention denoising module, a novel architecture for DDPMs. Fig. 2 shows an overview of the DiffCRN. In Fig. 2, the proposed model consists of two stages. In the first stage, i.e., spatial–spectral diffusion contrastive representation part, the unlabeled training data $\{\mathcal{T}_i, \mathcal{H}_0^i\}_{i=1}^N$ in the training set $\mathcal{T}$ are fed into DifCRN after forward diffusion process (6) and then estimate ϵ using network $\boldsymbol{\epsilon}_\theta(\mathcal{H}_t, t)$. According to (11), we can roughly reconstruct $\hat{\mathcal{H}}_0$, and then, the pairs $\{\mathcal{H}_0, \hat{\mathcal{H}}_0\}$ are regarded as two views and then fed into CL module $\mathscr{A}_\alpha$ for CL.

2) Spatial Self-Attention Denosing Module (SSAD): In addition to noisy hyperspectral instance embedding feature inputs, diffusion models sometimes process additional conditional information. Here, we consider the noise timesteps t. The noisy hyperspectral instance $\mathcal{H}_t \in \mathbb{R}^{C\times P\times P}$ is first embedded by a 2-D convolution layer with layer norm and Gaussian error linear unit (GELU) activation, which is denoted as $\mathscr{F}(\mathcal{H}_t^{C\times P\times P})$. For better denoising in the spatial domain and approximate information and distribution of spatial features, we introduce depthwise convolution and spatial self-attention (SSA) module into SSAD, as shown in Fig. 3. We can express this process using the following equation:

$$\begin{aligned}\mathcal{A} &= \mathrm{LN}\big(\mathrm{DWConv}\big(\mathscr{F}\big(\mathcal{H}_t^{C\times P\times P}\big)\big)\big)\\ \hat{\mathcal{A}} &= \mathcal{A}\cdot(1+\gamma)+\kappa\\ \hat{\mathcal{A}} &= \mathscr{F}\big(\mathcal{H}_t^{C\times P\times P}\big) + \mathrm{LayerScale}\cdot\mathrm{SSA}(\hat{\mathcal{A}})\end{aligned} \tag{12}$$

where γ and κ are the corresponding scale and shift based on the input timestep conditioning to modulate $\mathcal{A}$, which can be represented as

$$\gamma, \kappa = \mathsf{Chunk}(\mathrm{MLP}(\mathrm{TimeEmbeddings})) \tag{13}$$

where Chunk is splitting the feature into two parts of equal size along the channel dimension. MLP is implemented as a sigmoid linear unit (SiLU) activation followed by a linear layer. Sinusoidal TimeEmbeddings are created as follows:

$$\mathrm{TimeEmbeddings} = \left[\cos\left(\frac{t}{10\,000^{\frac{2i}{d}}}\right), \sin\left(\frac{t}{10\,000^{\frac{2i}{d}}}\right)\right] \tag{14}$$

where $t \in \mathbb{R}^B$, B represents the batch size of timestep, $i = 0, \ldots, (d/2) - 1$, and d represents the dimension of embedding. $[\cdot]$ represents the concatenation operation. SSA module computes self-attention within a local patch, as shown in Fig. 4. Then, SSA can be represented by

$$\mathrm{SSA}(\mathcal{Q}, \mathcal{K}, \mathcal{V}) = \mathrm{Attention}(\mathcal{Q}, \mathcal{K}, \mathcal{V}) = \sigma\left(\frac{\mathcal{Q}\mathcal{K}^T}{\sqrt{C}}\right)\mathcal{V} \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

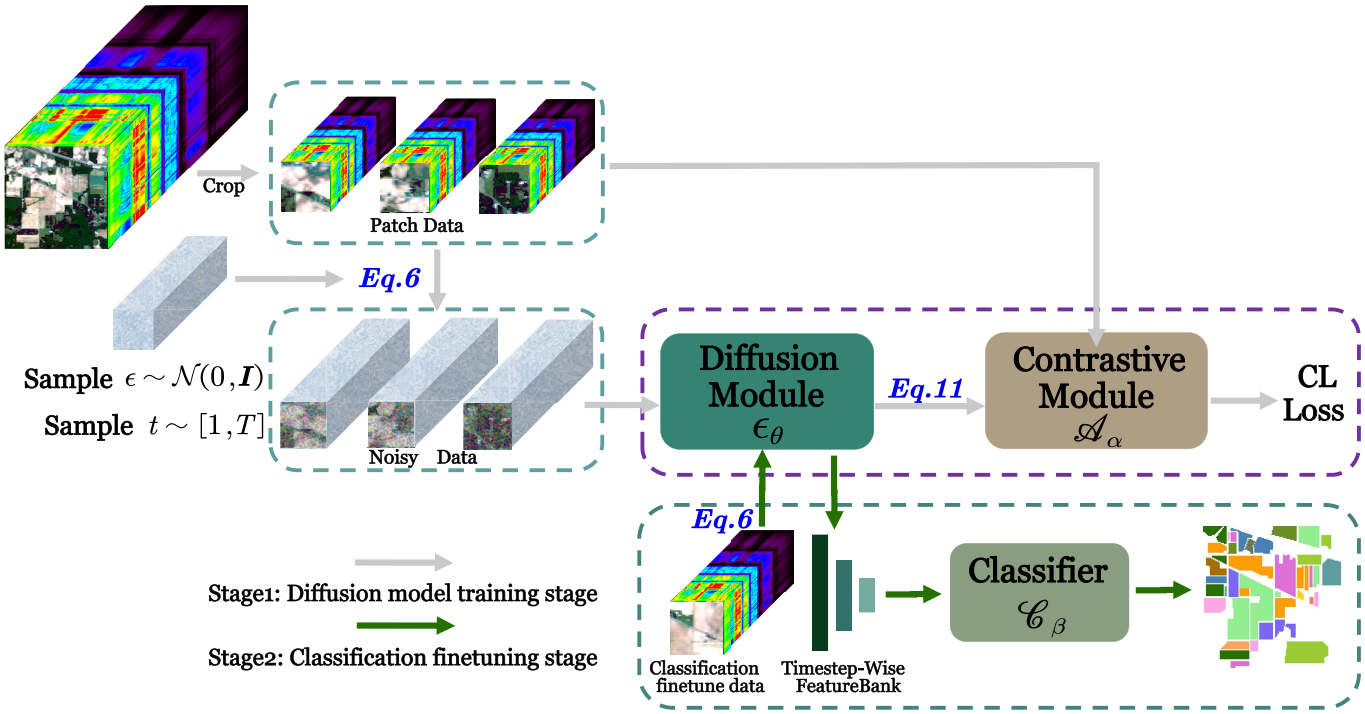


Fig. 2. Overall framework of the proposed DiffCRN model. The method consists of two steps. Step 1: as shown in the gray line, we pretrain the diffusion adversarial representation network ϵ_θ and $\mathscr{A}_\alpha$ with cropped HSI patches in an unsupervised manner. Step 2: as shown in the green line, another cropped HSI patch is fed into the pretrained diffusion adversarial representation network whose parameters are frozen, and then, the feature from different stages on specific timestep t is extracted. The extracted features on the same stage, but different t's are weighted and added. Finally, a classifier $\mathscr{C}_\beta$ is trained by using the five-timestep representations of limit-labeled data to perform HSI classification.

where $\mathcal{Q} = \mathcal{X}\mathcal{W}^Q, \mathcal{K} = \mathcal{X}\mathcal{W}^K, \mathcal{V} = \mathcal{X}\mathcal{W}^V \in \mathbb{R}^{C\times P\times P}$ are patch queries, keys, and values. $\mathcal{W}^Q$, $\mathcal{W}^K$, and $\mathcal{W}^V$ represent the projection weights for $\mathcal{Q}, \mathcal{K}, \mathcal{V}$, and $\mathcal{X}$, C represents the inputs and total number of spectral domain, respectively. σ indicates the softmax function.

3) Spectral Group Self-Attention Denoising (SGSAD) Module: The output of the second SSAD will first be fed into a 2-D convolution layer and then into the SGSAD. SGSAD is a well-designed block for better denoising in the spectral domain and approximating the information and distribution of spectral features. We introduce pointwise convolution and spectral group self-attention (SGSA) into SGSAD, as shown in Fig. 3. We can express this process using the following equation:

$$\begin{aligned}\mathcal{B} &= \mathrm{LN}\big(\mathrm{PWConv}\big(\mathscr{F}\big(\mathcal{S}_t^{C\times P\times P}\big)\big)\big)\\ \hat{\mathcal{B}} &= \mathcal{B}\cdot(1+\gamma)+\kappa\\ \hat{\mathcal{B}} &= \mathscr{F}(\mathcal{S}_t^{C\times P\times P}) + \mathrm{LayerScale}\cdot\mathrm{SGSA}(\hat{\mathcal{B}})\end{aligned} \tag{16}$$

where $\mathcal{S}_t^{C\times P\times P}$ is the output of the second spatial denoising module at t timestep. Fig. 3 indicates the detailed stage architecture of the spatial–spectral denoising network.

In SGSA, we apply attention mechanisms on the transpose of patch-level tokens, and the dimension change from $\mathbb{R}^{C\times P\times P}$ to $\mathbb{R}^{P^2\times C}$, P indicates the patch size. Thus, spectral tokens in SGSA interact with global spatial information on the spectral dimension. To reduce the computational complexity, we group the spectral domain into N_g groups and perform self-attention within each group, each group with dimension C_g. In this way, our spectral group attention is global, with patch-level tokens interacting across a group of channels, as shown in Fig. 4. Then, SGSA can be represented by

$$\mathrm{SGSA}(\mathcal{Q}, \mathcal{K}, \mathcal{V}) = [\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_{N_g}]$$

where

$$\mathrm{head}_i = \mathrm{SGSA}_{\mathrm{group}}(\mathcal{Q}_i, \mathcal{K}_i, \mathcal{V}_i) = \sigma\left(\frac{\mathcal{Q}_i\mathcal{K}_i^T}{\sqrt{C_g}}\right)\mathcal{V}_i \tag{17}$$

where $\mathcal{Q}_i, \mathcal{K}_i$, and $\mathcal{V}_i \in \mathbb{R}^{P^2\times C_g}$ are grouped channelwise patch-level queries, keys, and values.

4) CL Module: Fig. 5 shows the structure of the CL module in DiffCRN, which is comprised of six convolutional layers and a linear layer. Each convolutional layer includes a 2-D convolution layer with 3 × 3 convolutional kernels, a layer normalization, and a GELU activation. The number of kernels from the first to sixth convolution layers is correspondingly designed as 96, 96, 128, 128, 256, and 256. The CL module regularly learns spatial downsampling and increases the feature dimension. The inputs of CL module are reconstructed $\hat{\mathcal{H}}_0$ by (11) and raw hyperspectral instance $\mathcal{H}_0$. After feeding inputs into six convolutional layers, an average pooling (AvgPool) operation is adopted before a linear layer.

C. Spatial–Spectral DiffCRN Optimizing

Now, we need to define the form of the composite loss function, which contains three parts. Here, we explain as follows.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

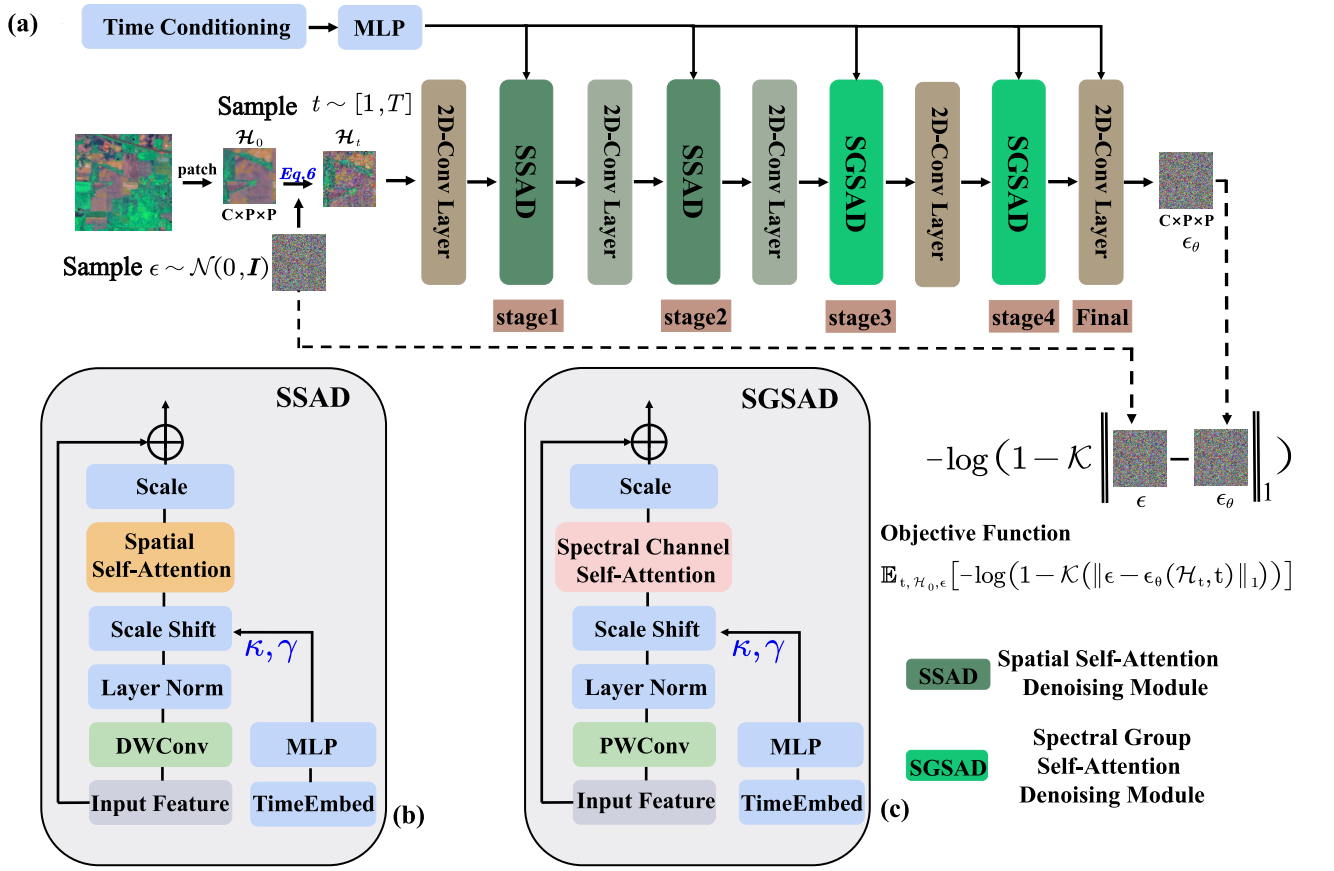


Fig. 3. Stage architecture of the spatial–spectral denoising adversarial representation learning network. Timestep embeddings are generated by an MLP. DWConv and PWConv represent depthwise convolution and pointwise convolution, respectively. (a) Network consists of two SSAD modules and two SGSAD modules. (b) Details of SSAD. (c) Details of the SGSAD module.

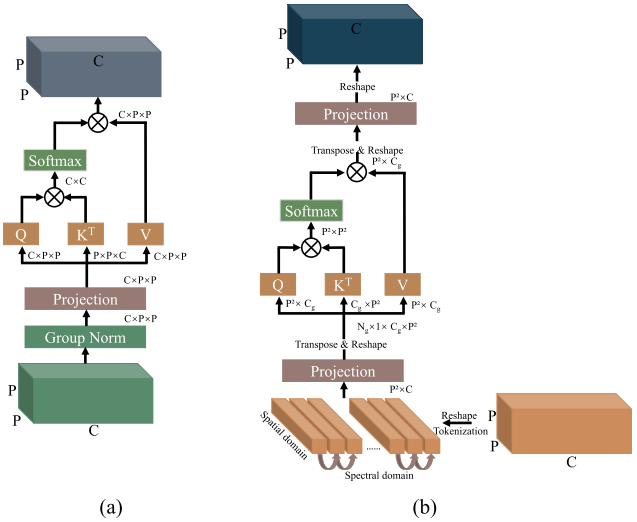


Fig. 4. (a) SSA module computes self-attention within the local patch. (b) SGSA module groups spectral tokens into multiple groups. Attention is performed in each spectral group with the entire patch-level spectrum as a token. In this work, we introduce SSA into SSAD to obtain local features and introduce SGSA into SGSAD to obtain global features. P, C, N_g, and C_g represent the patch size, the number of spectral domains, the number of spectral groups, and the spectra per group, respectively.

1) Spatial–Spectral Diffusion Loss: By maximizing its variational lower bound, it can turn into a simple supervised loss,

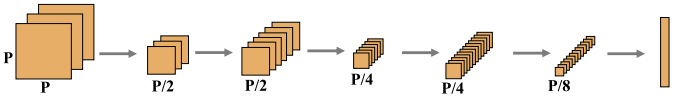


Fig. 5. CL module in DiffCRN.

i.e., mean squared error (mse), which is the preferred loss function in DDPMs [43], i.e., (18), also in some current works, including [52], [53], [54], [56], [58], [59], and [91]

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t,\mathcal{H}_0,\epsilon}\left[\|\epsilon - \epsilon_\theta(\mathcal{H}_t, t)\|^2\right] \tag{18}$$

where t is uniform between 1 and T, while the mse treats all the predicted noise uniformly. However, due to the variance schedule, the noise level in the diffusion forward process at an arbitrary timestep t is quite different. Inspired by [95], we introduce a nonuniform loss function for the diffusion model, named LAE; thus, we modify the (18) as follows:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t,\mathcal{H}_0,\epsilon}[-\log(1 - \mathcal{K}(\|\epsilon - \epsilon_\theta(\mathcal{H}_t, t)\|_1))] \tag{19}$$

where $\mathcal{K}$ is a clipping function, i.e., $\mathcal{K}(\cdot) \in [0, 0.9999)$. The reason for adopting such a logarithm loss function is to put more emphasis on heavily penalizing the big gap between ϵ and $\epsilon_\theta(\mathcal{H}_t, t)$. In the following ablation study, we verify the effectiveness of LAE.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

2) Reconstruction Loss: In view of the good properties of DDPMs in mathematical derivation, according to (11), we can reconstruct $\hat{\mathcal{H}}_0$ roughly. We use the similar form of (19) to measure the distance of $\hat{\mathcal{H}}_0$ and $\mathcal{H}_0$, which can be defined as

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{t,\mathcal{H}_0,\epsilon}\left[-\log\left(1-\mathcal{K}\left(\left\|\hat{\mathcal{H}}_0-\mathcal{H}_0\right\|_1\right)\right)\right]. \tag{20}$$

3) CL Loss: CL is a prominent approach in self-supervised learning that aims to learn useful representations from unlabeled data [96]. Accordingly, in DiffCRN, the contrastive loss function is designed to maximize similarity between fast-sampled fake hyperspectral imagery $\hat{\mathcal{H}}_0$ through (11) and the raw hyperspectral instance $\mathcal{H}_0$ while maximizing the difference between two distinct samples. Thus, $\hat{\mathcal{H}}_0 \in \mathbb{R}^{B\times C\times H\times W}$ and $\mathcal{H}_0 \in \mathbb{R}^{B\times C\times H\times W}$ are treated as two views for the given training sample. For the contrastive leaning task, $\hat{\mathcal{H}}_0$ and $\mathcal{H}_0$ are fed into the CL module to get the latent feature $\mathbf{F}'_{\hat{\mathcal{H}}_0} \in \mathbb{R}^{B\times 256}$ and $\mathbf{F}_{\mathcal{H}_0} \in \mathbb{R}^{B\times 256}$. A particularly successful contrastive loss, named InfoNCE, is adopted [96]. It takes the following form:

$$\begin{aligned}\mathcal{L}_{\text{con}} &= \frac{1}{2B}\sum_{k=1}^{B}\left(\ell(2k-1,2k)+\ell(2k,2k-1)\right)\\ \ell(i,j) &= -\log\frac{e^{\text{sim}\left(\mathbf{F}'^{[i]}_{\hat{\mathcal{H}}_0},\mathbf{F}^{[j]}_{\mathcal{H}_0}\right)/\tau}}{\Sigma_{k=1}^{2B}\mathbb{1}_{[k\neq i]}e^{\text{sim}\left(\mathbf{F}'^{[i]}_{\hat{\mathcal{H}}_0},\mathbf{F}^{[k]}_{\mathcal{H}_0}\right)/\tau}}\end{aligned} \tag{21}$$

where $\mathbf{F}'^{[i]}_{\hat{\mathcal{H}}_0}$ and $\mathbf{F}^{[j]}_{\mathcal{H}_0}$ are two views for the ith and jth sample. Here, sim: $\mathbb{R}^d \rightarrow \mathbb{R}$ is a similarity metric (e.g., cosine similarity). τ is a hyperparameter that controls the magnitude of the loss. B is the number of training samples in one minibatch, and $\mathbb{1}_{[k\neq i]}$ is an indicator function: 1 if $k \neq i$ and 0 otherwise.

4) Compound Loss Function: It is obvious that the proposed framework DiffCRN is a multitask learning network. It is necessary to define a weighted sum of these four losses considering the contributions of different loss components rather than using predefined or manually tuned weights. Inspired by [97], we define the compound loss based on the homoscedastic uncertainty of each task, which can learn various quantities and units in each task. We define it as follows:

$$\mathcal{L} = e^{-w_{\text{diff}}}\mathcal{L}_{\text{diff}} + e^{-w_{\text{rec}}}\mathcal{L}_{\text{rec}} + 0.5\mathcal{L}_{\text{con}} + w_{\text{diff}} + w_{\text{rec}} \tag{22}$$

where $w_{\text{diff}} = \log v_{\text{diff}}$, $w_{\text{rec}} = \log v_{\text{rec}}$, v_{diff}, and v_{rec} are the noise variance representing the uncertainty of the reconstruction task of (19) and (20). During pretraining of the spatial–spectral diffusion model, we set w_{diff} and w_{rec} as the learning parameters of the model and initialize them to 0. The last two terms w_{diff} and w_{rec} act as a regularizer to avoid overfitting. Notably, Fig. 6 shows the changes of w_{diff} and w_{rec} on specific dataset, i.e., task uncertainty. We can see that as the increasing of epoch, w_{diff} and w_{rec} gradually become smaller and converges, which means that the weight of $\mathcal{L}_{\text{diff}}$ and $\mathcal{L}_{\text{rec}}$ becomes bigger, resulting the main optimization object.

5) Pixel-Level SAM: Unlike [54], [59], which takes feature representations of the diffusion model at the manually tuned timesteps, then feeds them into the classifier, and Zhou et al. [60] use all of the timesteps, which might lead to redundancy.

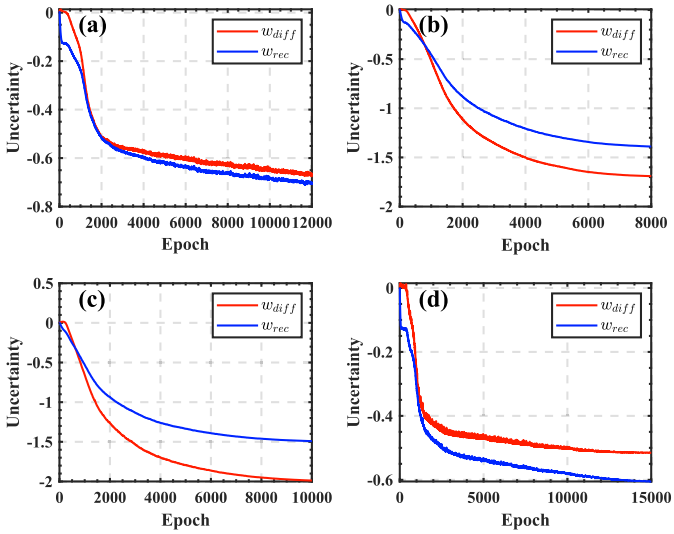


Fig. 6. Task uncertainty over spatial–spectral diffusion process loss (red) and reconstruction loss (blue). The task uncertainty converges as the number of epochs increases. (a) Indian pines. (b) PU. (c) MUUFL (d) WHU-Hi-HongHu.

For dynamic adjustment and decision, it is essential to adjust which timestep should be used iteratively during the phase of training. Hence, we propose a simple but efficient methodology based on learnable SAM to measure the similarity on the pixel level, given as

$$\mathbf{SAM}(\hat{\mathcal{H}}_0,\mathcal{H}_0) = \cos^{-1}\left(\frac{\langle\hat{\mathcal{H}}_0,\mathcal{H}_0\rangle}{\left\|\hat{\mathcal{H}}_0\right\|_2\cdot\|\mathcal{H}_0\|_2}\right) \tag{23}$$

where $\hat{\mathcal{H}}_0 \in \mathbb{R}^{B\times C\times P\times P}$, $\mathcal{H}_0 \in \mathbb{R}^{B\times C\times P\times P}$, and $\mathbf{SAM} \in \mathbb{R}^{B\times P\times P}$; here, B is the number of hyperspectral instance, each instance corresponds to a timestep t. The smaller the SAM, the higher the similarity. The mean value of SAM can be obtained along the B dimension. In the following experiments, we take the timestep corresponding to the top five most similar of SAM after averaging. In the following ablation study, we verify the effectiveness of SAM.

D. Classification With Spectral–Spatial Diffusion Features

1) Extracting Intermediate Representations From DiffCRN: Given a hyperspectral instance, we first extract the intermediate feature representations from the pretrained spectral–spatial diffusion model and feed these features as the inputs to the classifier $\mathscr{C}_\beta$, as shown in Fig. 7. To achieve multilevel representation, we retain the features from five different timesteps t, which are selected from the top five smallest values of **SAM**. The above process can be expressed as

$$\begin{aligned}\chi_i &= \text{Conv2d}(\mathcal{F}(\epsilon_\theta(\mathcal{H}_t,t)))\\ \tilde{\psi} &= \text{CTSSFM}\left(\sum_{s=1}^{5}\text{AWAM}\left(\sum_{j=1}^{5}\chi_j\right)\right)\\ \hat{y} &= \text{FC}(\text{GAP}(\tilde{\psi}))\end{aligned} \tag{24}$$

where $\mathcal{F}(\cdot)$ is the extracted feature representations on different timestep t. $\tilde{\psi}$ means the concatenated features. s represents the stage index in DiffCRN, as shown in Fig. 3 (i.e., stage1, stage2, stage3, stage4, and final), and j represents the jth time

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

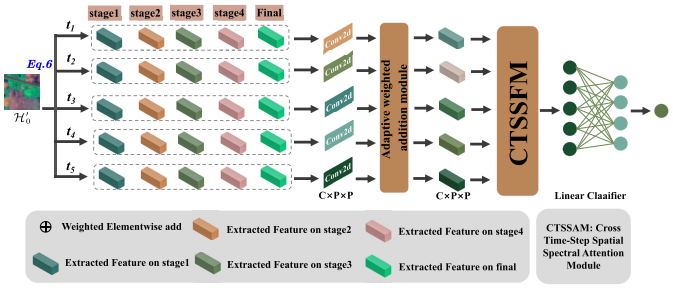


Fig. 7. Classifier with spectral–spatial diffusion feature as inputs. The input data of the classifier are the extracted features from the diffusion model on a specific timestep; here, we choose five timesteps. First, a convolutional layer with a kernel size of 1 is applied to the extracted features to change to dimension. Then, the feature maps are fed into AWAM and CTSSFM, followed by a linear classifier to perform HSIC.

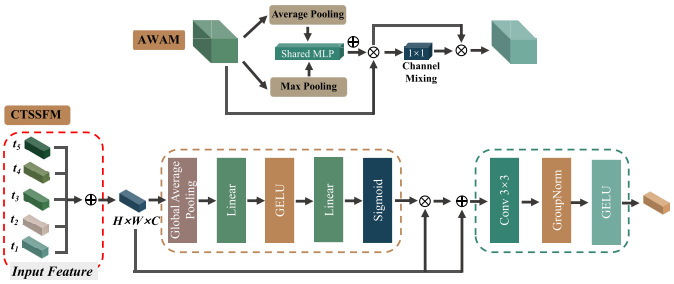


Fig. 8. (Top): AWAM. The input data of AWAM is the sum of extracted features on the same stage but different timesteps after Conv1 × 1. (Bottom): CTSSFM. The input feature of CTSSFM is the elementwise summation of the output of AWAM.

index. $\hat{y}$ represents the predicted label corresponding to the hyperspectral instance.

2) Adaptive Weighted Addition Module (AWAM): The AWAM performs channel attention by exploring the inter-channel relationships present within the features, as shown in Fig. 8. Given a feature map $F \in \mathbb{R}^{C\times H\times W}$, which is the elementwise addition of extracted feature maps from the same stage but different diffusion timesteps, as input, AWAM infers 1-D channel attention map $M(F) \in \mathbb{R}^{C\times 1\times 1}$. The refined feature map $F' \in \mathbb{R}^{C\times H\times W}$ is computed as

$$F' = (F \otimes M(F)) \otimes \mathrm{Conv}_{1\times 1}(F \otimes M(F)) \tag{25}$$

where $\otimes$ denotes elementwise multiplication. In $M(F)$, AvgPool and max pooling (MaxPool) operations are employed to aggregate spatial information. Subsequently, features are fed into a shared multilayer perceptron (MLP) activated by the sigmoid function. The feature map is obtained by combining the outputs of the shared MLP through elementwise summation. Then, the 1-D channel attention map $M(F) \in \mathbb{R}^{C\times 1\times 1}$ is broadcast along the spatial dimension to obtain the refined feature. The 1-D channel attention map $M(F) \in \mathbb{R}^{C\times 1\times 1}$ is calculated as follows:

$$M(F) = \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F))). \tag{26}$$

3) Cross-Timestep Spectral–Spatial Fusion Module (CTSSFM): The detailed structure of CTSSFM is illustrated in Fig. 8. The input of CTSSFM is the elementwise summation of refined feature maps from different stages and diffusion timesteps, which differs from AWAM. We use $F \in \mathbb{R}^{C\times H\times W}$ to represent the sum of five given feature maps $F_i \in \mathbb{R}^{C\times H\times W}, i \in 1,\ldots,5$. CTSSFM performs channel attention and spatial attention sequentially, and then, a 3-D attention map $M(F) \in \mathbb{R}^{C\times H\times W}$ is refined as follows:

$$M(F) = \mathscr{F}_{\mathrm{spa}}(F \oplus (\mathscr{F}_{\mathrm{spe}}(F) \otimes F)) \tag{27}$$

where $\oplus$ denotes the elementwise summation. We use the global average pooling (GAP) in the spectral attention module $\mathscr{F}_{\mathrm{spe}}$ to get a feature descriptor on the global spatial information. Subsequently, a 1-D convolution module is used to extract the channels of interest. Finally, we get the spectral attention vector from a sigmoid function. In summary, $\mathscr{F}_{\mathrm{spe}}$ is calculated as follows:

$$\mathscr{F}_{\mathrm{spe}} = \sigma(\mathrm{GAP}(\mathrm{Conv}_{1\times 1}(F))). \tag{28}$$

In the spatial attention module $\mathscr{F}_{\mathrm{spa}}$, first, we apply a 3×3 convolution to maintain a large receptive field to aggregate the spatial context information, and then, a GroupNorm and GELU activation function are used. The calculation process is expressed as follows:

$$\mathscr{F}_{\mathrm{spa}} = \mathrm{GELU}(\mathrm{GruopNorm}(\mathrm{Conv}_{3\times 3}(*)) \tag{29}$$

where * is the output of $\mathscr{F}_{\mathrm{spe}}$. In the following ablation study, we verify the effectiveness of AWAM and CTSSFM.

4) Classification Loss: In the second stage, i.e., classification task, we adopt the cross-entropy loss, which can be written as

$$\mathcal{L}_{\mathrm{cls}} = \frac{1}{N}\sum_{i=1}^{N}[y_i \log \mathscr{C}_\beta(\mathcal{F}(\epsilon_\theta(\mathcal{H}_t, t))) - (1-y_i)\log \mathscr{C}_\beta(1-\mathcal{F}(\epsilon_\theta(\mathcal{H}_t, t)))] \tag{30}$$

where N denotes the number of labeled cubes and y_i denotes the class label of the ith cube in classification training set $\mathcal{L}$.

IV. Experimental Results and Analysis

A. Datasets' Description

To evaluate the performance of the proposed method, four classical HSI datasets are adopted, i.e., Indian Pines (IN),[1] Pavia University (PU),[2] WHU-Hi-HongHu,[3] and MUUFL.[4] The class name and the number of training and testing are listed in Table I. Exp1 (10%, 3%, 0.5%, and 3% training samples for IN, PU, WHU-Hi-HongHu, and MUUFL, respectively) and Exp2 represent different sample selections. Fig. 9 shows the false-color map and ground-truth map.

1) Indian Pines Data: This dataset was collected by the AVIRIS sensor over Northwestern Indiana, USA. These data consist of 145 × 145 pixels at a ground sampling distance (GSD) of 20 m and 220 spectral bands covering the wavelength range of 400–2500 nm with a 10-m spectral resolution. In the experiment, 24 water-absorption bands and noise bands were removed, and 200 bands were selected. There are 16 main investigated categories in this studied scene.

[1]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[2]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[3]http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

[4]https://github.com/GatorSense/MUUFLGulfport

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I

LAND-COVER TYPES, AND TRAINING AND TESTING SAMPLE NUMBERS OF THE INDIAN PINES DATASET, THE PU DATASET, THE WHU-HI-HONGHU DATASET, AND THE MUUFL DATASET

No.	Indian Pines			Pavia University			WHU-Hi-HongHu			MUUFL		
	Class	Training Exp1/Exp2[1]	Testing Exp1/Exp2	Class	Training Exp1/Exp2	Testing Exp1/Exp2	Class	Training Exp1/Exp2	Testing Exp1/Exp2	Class	Training Exp1/Exp2	Testing Exp1/Exp2
1	Alfalfa	5/20	41/26	Asphlat	199/20	6432/6611	Red roof	70/50	13971/13991	Trees	697/20	22549/23226
2	Corn-notill	143/20	1285/1408	Meadows	559/20	18090/18629	Road	18/50	3494/3462	Mostlt grass	128/20	4142/4250
3	Corn-mintill	83/20	747/810	Gravel	63/20	2036/2079	Bare soil	109/50	21712/21771	Mixed ground surface	206/20	6676/6862
4	Corn	24/20	213/217	Trees	92/20	2972/3044	Cotton	816/50	162469/163235	Dirt and sand	55/20	1771/1806
5	Grass-pasture	48/20	435/463	Painted metal sheets	40/20	1305/1325	Cotton firewood	31/50	61876168	Road	201/20	6486/6667
6	Grass-trees	73/20	657/710	Bare Soil	151/20	4878/5009	Rape	223/50	44334/44507	Water	14/20	452/446
7	Grass-pasture-mowed	3/20	25/8	Bitumen	40/20	1290/1310	Chinese cabbage	121/50	23982/24053	Building Shadow	67/20	2166/2213
8	Hay-windrowed	48/20	430/458	Self-Blocking Bricks	110/20	3572/3662	Pakchoi	20/50	4034/4004	Building	187/20	6053/6220
9	Oats	2/15	18/5	Shadows	28/20	919/972	Cabbage	54/50	10765/10769	Sidewalk	42/20	1343/1365
10	Soybean-notill	97/20	875/952				Tuber mustard	62/50	12332/8904	Yellow curb	5/20	178/163
11	Soybean-mintill	246/20	2209/2435				Brassica parachinensis	55/50	10960/10965	Cloth panels	8/20	261/249
12	Soybean-clean	59/20	534/573				Brassica chinensis	45/50	8909/8904			
13	Wheat	21/20	184/185				Small Brassica chinensis	113/50	22394/22457			
14	Woods	127/20	1138/1245				Lactuca sativa	37/50	7319/7212			
15	Building-Grass-Trees-Drives	39/20	347/366				Celtuce	5/50	997/952			
16	Stone-Steel-Towers	9/20	84/73				Film covered lettuce	36/50	7226/7212			
17							Romaine lettuce	15/50	2995/2960			
18							Carrot	16/50	3201/3167			
19							White radish	44/50	8668/8662			
20							Garlic sprout	17/50	3469/3436			
21							Broad bean	7/50	1321/1278			
22							Tree	20/50	4020/3990			
	Total	1027/315	9222/9934		1282/180	41494/42596		1934/1100	384759/385593		1610/220	52077/53467

[1] EXP1/EXP2 means two experiments with different number of training samples. In EXP2, we choose SSRN, MTGAN, SSFTT, and SC-SS-MTr as comparative methods.

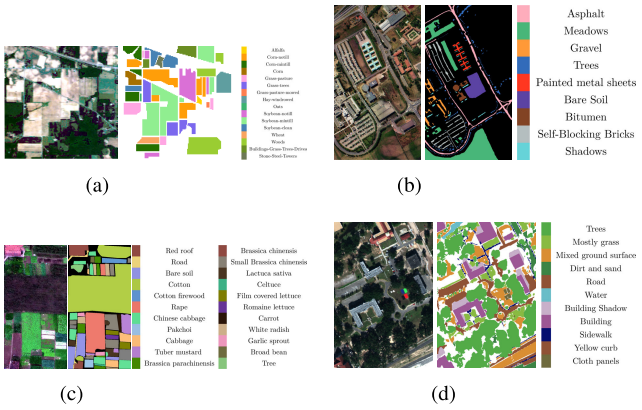


Fig. 9. Pseudocolor image, the corresponding ground-truth map, and the legend of the four datasets. (a) Indian Pines. (b) PU. (c) WHU-Hi-HongHu. (d) MUUFL.

2) PU Data: It was acquired by the reflective optics system imaging spectrometer (ROSIS) sensor over PU and its surroundings, Pavia, Italy. This dataset has 103 spectral bands ranging from 430 to 860 nm. Its spatial resolution is 1.3 m, and its image size is 610 × 340. Nine land-cover categories are covered.

3) WHU-Hi-HongHu Data: This dataset was acquired on November 20, 2017, by Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform over the area of Honghu City, Hubei, China, with a spatial resolution of 0.043 m, an image size of 940 × 475, and 270 bands in the range of from 400 to 1000 nm. Twenty-two land-cover categories are covered.

4) MUUFL Data: It was obtained in November 2010 around the region of the University of Southern Mississippi Gulf Park, Long Beach, MS, USA, by using the ROSIS sensor. It is made up of 325 × 220 pixels along with 72 spectral bands. The first and last eight bands are deleted owing to noise, resulting in 64 bands in total. There are 53 687 ground-truth pixels with 11 different types of classes for urban land cover.

B. Experimental Settings

1) Implementation Details and Training Details: Our proposed DiffCRN model is implemented on the PyTorch 1.10.2 platform using a workstation with Intel[5] Xeon[5] CPU E5-2640 v4, 256-GB RAM, and an NVIDIA GeForce RTX 2080 Ti 11-GB GPU. Training and testing samples are extracted as 3-D cubes. The Adam optimizer is adopted with a learning rate of $1e^{-4}$. In the pretraining stage, we set the batch size of 1024, 640, 480, and 640 and the patch size of 7, 9, 11, and 9 for Indian Pines, PU, WHU-Hi-HongHu, and MUUFL datasets, respectively. The number of epochs is set to 12 000, 8000, 15 000, and 15 000 for these datasets, respectively. In the classification stage, we set the batch size of 64, 64, 128, and 128 for the four datasets, respectively. The number of epochs is set to 300 for all datasets. We calculate the results by averaging the results of ten repeated experiments with different training samples selected randomly.

2) Comparing With Other Methods: In order to compare the effectiveness of the proposed DiffCRN, several representative algorithms were selected for the control experiments, including three conventional classifiers (i.e., support vector machine (SVM), k-nearest neighbors (KNN), and RF), CNN-based networks (i.e., 1-D CNN [98], 2-D CNN [77], 3-D CNN (TGRS2018) [15], SSRN (TGRS2018) [16], HybridSN (GRSL2020) [13], and SS-ConvNeXt (JSTARS2023) [99]), GAN-based networks (i.e., MTGAN (TGRS2020) [39] and ADGAN (TGRS2021) [100]), transformer-based networks (i.e., SSFTT (TGRS2022) [82], SSTN (TGRS2022) [81]), and unsupervised method (i.e., SC-SS-MTr (TGRS2023) [101]), and DDPM based methods (i.e., DEMAE (TGRS2024) [64] and DKDMN (ESWA2024) [63]). All the parameter settings are the same in the corresponding reference.

1) The SSRN introduces the ResNet structure into the designed spatial 3-D CNN module and spectral 3-D CNN module and extracts rich spatial and spectral features.

[5]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, k, FWIOU, MIOU, AND THE ACCURACIES FOR EACH CLASS ON THE INDIAN PINES DATASET WITH 10% TRAINING SAMPLE. THE BEST RESULTS ARE COLORED SHADOW AND OPTIMAL STANDARD DEVIATION IN BOLD. RESULTS ON EXP1

Class No.	Color	Conventional Classifier			CNNs Based networks						GAN Based Networks		Transformer Based Networks		Unsupervised Network	Diffusion Model Based		DiffCRN
		KNN	RF	SVM	1D-CNN [104]	2D-CNN [78]	3D-CNN [80]	HybridSN [81]	SSRN [82]	SS-ConvNeXt [105]	MTGAN [39]	ADGAN [106]	SSFTT [86]	SSTN [85]	SC-SS-MTr [107]	DEMAE [64]	DKDMN [63]	
1		11.43±9.38	7.86±9.12	15.24±6.85	58.78±29.66	76.59±10.36	81.71±18.27	88.78±5.66	90.98±9.13	86.05±12.16	84.63±9.89	97.87±2.75	94.18±7.55	94.39±5.02	91.95±6.82	97.86±4.19	88.78±11.77	98.65±**1.30**
2		57.17±4.36	64.70±1.92	67.15±2.99	64.79±6.14	86.05±5.22	86.89±2.94	95.60±1.46	97.54±1.02	97.88±1.24	97.20±1.40	96.33±1.49	94.90±2.99	98.01±1.27	97.28±1.35	96.98±1.29	92.63±6.79	99.36±**0.32**
3		45.18±2.13	54.61±2.20	55.07±2.69	57.40±8.06	84.99±3.99	81.97±4.44	94.04±2.32	96.88±2.48	98.54±2.06	95.90±3.13	97.71±2.95	92.16±4.24	92.05±5.24	97.54±1.35	98.81±1.20	95.93±3.77	99.01±**0.90**
4		19.02±4.49	27.99±7.99	34.72±5.36	58.08±8.49	85.31±7.80	84.90±5.00	90.52±6.67	97.70±2.46	99.38±**0.75**	96.85±2.77	98.36±1.76	94.80±3.00	96.24±2.30	94.98±3.60	97.94±2.00	86.09±8.28	99.13±1.19
5		81.45±2.39	83.06±4.11	78.85±4.07	72.87±7.95	82.99±4.71	90.59±3.86	95.01±2.32	95.13±2.78	94.53±2.65	95.61±3.42	95.11±4.72	95.33±2.92	94.34±4.11	94.29±2.32	97.45±1.81	91.49±2.52	98.30±**1.38**
6		96.01±1.52	95.65±1.69	94.67±1.99	90.20±5.28	95.28±2.51	95.35±4.16	99.60±0.39	99.25±0.74	98.82±0.67	98.48±1.12	97.34±2.59	96.66±3.27	98.66±0.66	99.56±0.53	99.57±0.45	96.67±4.04	99.87±**0.14**
7		36.15±24.84	2.31±**3.72**	9.62±5.51	64.00±23.40	67.20±17.67	82.67±14.10	95.20±4.92	76.40±32.25	95.45±5.67	12.00±29.45	85.50±12.46	84.92±22.10	92.80±11.12	36.00±32.10	92.69±13.48	84.00±11.66	98.29±4.54
8		95.54±1.23	98.10±0.97	98.35±1.05	94.28±5.69	98.35±0.96	99.46±1.12	99.70±0.69	99.53±1.47	99.98±0.07	99.95±0.10	96.62±0.53	99.62±0.56	99.95±0.15	99.95±0.09	99.84±0.42	100.00±**0.00**	100.00±**0.00**
9		6.11±11.84	6.11±8.05	11.11±9.80	63.53±20.15	74.71±20.94	74.02±18.17	91.18±5.72	55.29±39.24	90.71±11.19	61.76±43.87	83.82±17.72	77.83±16.66	93.53±7.04	8.89±7.11	97.78±3.69	85.78±17.07	99.16±**2.22**
10		69.39±3.39	65.82±5.49	62.69±2.67	67.79±10.40	84.41±3.79	79.03±4.41	94.31±2.23	96.19±1.52	97.12±1.60	95.43±2.60	95.21±2.63	92.21±4.95	96.02±1.86	95.71±1.82	96.15±1.95	91.59±4.10	98.01±**0.69**
11		70.10±1.99	86.57±1.13	74.03±1.69	79.17±8.07	90.27±1.95	88.15±3.56	95.92±2.43	98.29±1.14	98.95±0.69	98.44±1.10	98.76±1.07	97.52±1.25	98.11±1.11	99.48±**0.19**	99.02±0.57	97.96±2.32	98.43±0.47
12		28.03±5.95	45.58±4.98	49.89±3.09	42.34±10.33	76.07±4.86	79.59±4.99	93.63±1.98	97.97±1.70	97.94±1.87	95.10±3.07	98.46±2.49	90.48±9.01	97.70±1.56	96.53±1.37	94.64±2.74	94.64±2.74	99.30±**0.66**
13		81.41±2.53	92.81±1.82	92.16±3.41	95.41±2.89	94.86±6.67	93.33±5.68	99.73±0.58	99.68±0.58	99.62±0.64	98.76±2.18	98.24±1.65	96.59±6.15	99.84±0.51	97.83±3.77	97.49±2.52	97.48±2.52	99.85±**0.41**
14		92.92±1.80	95.21±2.22	89.51±3.15	94.47±3.10	96.77±1.72	96.11±1.82	99.32±0.53	99.57±0.43	98.43±3.24	99.12±1.62	99.89±**0.10**	98.67±1.47	99.37±0.59	99.68±0.36	99.80±0.30	99.23±0.65	99.58±0.37
15		15.14±3.11	39.28±5.99	52.24±6.96	44.24±10.57	86.97±8.85	81.53±9.74	95.73±2.66	98.01±2.61	97.99±2.39	98.10±1.56	97.19±3.82	83.48±12.18	90.61±9.10	98.47±1.26	99.28±0.87	95.33±3.67	99.79±**0.43**
16		80.00±2.62	79.05±5.92	69.76±8.77	91.67±3.55	97.02±3.38	81.45±7.79	98.45±2.69	97.38±2.68	97.65±3.70	92.50±5.24	88.39±11.57	89.93±15.59	97.26±1.78	70.60±13.97	97.50±2.94	99.76±**0.53**	99.66±0.58
AA (%) ↑		56.19±1.88	59.04±1.38	59.69±1.27	71.19±2.76	86.12±6.59	86.05±2.24	95.42±0.92	93.49±4.01	96.82±1.30	88.80±3.60	95.48±1.55	92.46±3.65	96.18±1.43	96.13±2.31	97.98±1.00	93.59±2.58	99.26±**0.37**
OA (%) ↑		68.20±0.76	74.85±1.00	74.63±0.48	73.75±0.80	88.83±1.57	87.90±1.11	96.18±0.76	97.81±0.66	98.20±0.52	97.20±0.50	97.63±0.90	95.12±1.76	97.15±0.88	97.33±0.40	98.43±0.36	95.84±1.36	99.33±**0.16**
k*100 ↑		63.52±0.88	73.85±1.07	70.93±0.55	70.52±0.81	88.40±1.63	87.34±1.21	96.07±0.81	97.74±0.69	98.11±0.58	97.08±0.52	97.94±0.85	94.79±1.95	96.94±0.96	96.95±0.46	98.21±0.41	95.4±1.56	99.31±**0.16**
FWIoU (%) ↑		30.92±2.33	41.80±2.13	41.82±2.06	60.07±0.85	81.91±2.27	80.52±1.64	93.44±1.28	96.15±1.14	96.81±0.93	95.05±0.85	96.48±1.39	91.57±3.02	95.06±1.37	95.30±0.75	96.95±0.68	92.03±2.65	98.81±**0.28**
MIoU (%) ↑		34.48±2.22	43.11±1.71	43.65±1.65	59.35±2.09	79.46±2.48	77.69±2.12	91.58±1.52	91.88±4.45	94.88±1.89	86.72±3.74	94.72±1.76	87.76±4.63	92.54±2.52	84.55±2.83	95.90±1.43	89.59±2.36	98.60±**0.55**

2) The HybridSN consists of 3-D CNN and 2-D CNN layers. An image patch sample passes through three 3-D CNN layers and one 2-D CNN layer successively to obtain a spectral–spatial joint feature.
3) The SS-ConvNeXt is our previous work published in 2023. A spectral-ConvNext and a Spatial-ConvNeXt block are designed to capture spectral and spatial information, respectively.
4) The MTGAN uses an encoder-decoder to generate fake images and a discriminator to discriminate the input sample coming from the real distribution or the reconstructed one. The extracted features in the encoder–decoder are used to make classification.
5) The ADGAN consists of a generator and a discriminator. The generator consists of four 2-D transposed convolution blocks. The discriminator consists of five 2-D CNN blocks and a softmax layer.
6) The SSFTT is a spatial–spectral transformer that utilizes the advantages of a hybrid CNN and transformer with a Gaussian-weighted feature tokenizer module.
7) The SSTN is a spectral–spatial transformer network, which consists of spatial attention and spectral association modules, to overcome the constraints of convolution kernels.
8) The SC-SS-MTr is a spectral–spatial masked transformer with supervised and CL, which pretrains a vanilla transformer via reconstruction from masked inputs and then fine-tunes for HSIC.
9) The DEMAE, a two-stage model, combines masked AE (MAE) and diffusion model to capture the spatial correlation and the data distribution under the condition of the category label.
10) The DKDMN establishes two views—one is the original hypersepctral patches and another is the reconstructed land-cover distribution knowledge—and uses CL to enhance the network's cross-sample awareness ability.

3) Evaluation Metrics: To quantitatively evaluate the proposed method and other compared methods, we choose following commonly used metrics, i.e., overall classification accuracy (***OA*** ↑), average classification accuracy (***AA*** ↑), category accuracy (***CA*** ↑), Kappa coefficient (k ↑), mean intersection over union (***MIoU*** ↑), frequency weighted intersection over union (***FWIoU*** ↑), and standard deviation (σ) of classwise accuracy. Among them, the MIoU and FWIoU can be computed by

$$\begin{cases} \text{MIoU} = \dfrac{1}{K+1}\sum_{i=0}^{K}\dfrac{N_{ii}}{\Sigma_{j=0}^{K}N_{ij}+\Sigma_{j=0}^{K}N_{ji}-N_{ii}} \\ \text{FWIoU} = \dfrac{1}{\Sigma_{i=0}^{K}\Sigma_{j=0}^{K}}N_{ij}\Sigma_{i=0}^{K}\dfrac{N_{ii}}{\Sigma_{j=0}^{K}N_{ij}+\Sigma_{j=0}^{K}N_{ji}-N_{ii}} \end{cases} \tag{31}$$

where $N \in \mathbb{R}^{K\times K}$ is the confusion matrix. ***MIoU*** is used to compare the similarity between the model prediction results and the true labels. ***FWIoU*** is the weighted IoU. Each experiment was repeated ten times, and the results were averaged.

C. Performance Analysis

1) Numerical Evaluation: Quantitative assessments of classification performance are presented from Tables II–V. The best classification values are colored shadow and optimal standard deviation displayed in bold. Our proposed DiffCRN introduces the diffusion model to HSI classification, considering extracted features from various diffusion timesteps t based on SAM, resulting in a powerful ability to model complex relations. The results show that the proposed DiffCRN is superior to all other techniques in terms of OA, AA, k, FWIoU, and MIoU and represents better performance on classwise accuracy simultaneously.

a) Performance on Indian Pines: Table II shows the quantitative comparison results of the Indian Pines dataset. Benefiting from the pretrained DiffCRN, the performance of DiffCRN is higher than other conventional classifiers, classic backbone, GAN-based networks, and transformered methods, as well as an unsupervised network with pretraining. For instance, DiffCRN is 6.7%, 11.8%, 3.9%, 10.8%, 6.1%, and 14.1% better than SSRN, MTGAN, ADGAN, SSFTT, SSTN, and SC-SS-MTr in terms of MIoU. The results show that DiffCRN can obtain label-domain classification results with higher precision masks that have better raster reversibility. In terms of AA, HybridSN, and SSRN provide better performance than a conventional classifier,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, k, FWIOU, MIOU, AND THE ACCURACIES FOR EACH CLASS ON THE PU DATASET WITH 3% TRAINING SAMPLE. THE BEST RESULTS ARE COLORED SHADOW AND OPTIMAL STANDARD DEVIATION IN BOLD. RESULTS ON EXP1

Class No.	Color	Conventional Classifier			CNNs Based networks						GAN Based Networks		Transformer Based Networks		Unsupervised Network	Diffusion Model Based		DiffCRN
		KNN	RF	SVM	1D-CNN [104]	2D-CNN [78]	3D-CNN [80]	HybridSN [81]	SSRN [82]	SS-ConvNeXt [105]	MTGAN [39]	ADGAN [106]	SSFTT [86]	SSTN [85]	SC-SS-MTr [107]	DEMAE [64]	DKDMN [63]	
1		84.90±1.81	89.04±1.40	88.79±1.77	84.17±3.93	95.30±1.26	95.08±0.96	96.22±3.78	98.74±0.63	98.79±1.60	98.68±0.75	71.35±18.54	86.53±29.32	98.20±0.96	99.35±0.44	99.77±0.40	99.71±0.71	99.83±**0.20**
2		96.35±1.52	96.93±0.60	95.65±0.61	95.48±1.83	98.09±0.70	98.63±0.49	99.22±0.68	99.74±0.15	99.73±0.43	99.79±0.12	94.08±5.23	94.84±13.20	96.58±1.52	99.92±0.07	99.98±0.05	99.13±0.39	99.98±**0.02**
3		57.30±2.62	53.19±4.16	68.54±3.25	59.47±11.73	77.83±4.32	86.72±5.28	84.33±6.05	90.69±5.71	94.66±2.77	97.16±2.89	65.31±50.62	87.57±28.90	97.23±1.61	96.76±1.43	94.95±1.89	97.51±1.32	98.64±**1.06**
4		75.50±3.96	84.09±2.94	85.13±3.30	85.90±4.41	97.97±0.58	95.44±1.42	98.07±1.07	97.30±0.82	96.21±0.94	97.44±**0.23**	58.87±22.54	84.86±30.53	96.09±0.68	97.40±1.29	98.77±1.89	98.30±0.99	99.17±0.38
5		99.03±0.32	97.62±0.44	91.38±2.76	99.10±0.83	99.93±0.14	99.36±1.54	100.00±**0.00**	99.89±0.15	99.86±0.13	99.59±0.30	92.82±11.32	99.39±1.22	99.92±0.12	99.75±0.38	100.00±**0.00**	99.84±0.15	100.00±**0.00**
6		45.92±4.74	51.65±1.99	82.67±2.47	50.62±7.93	91.17±1.76	89.67±3.75	95.58±1.95	99.50±0.49	99.70±0.41	99.73±0.21	84.95±14.71	79.68±41.77	99.51±1.08	99.84±0.23	99.99±**0.02**	99.90±0.12	99.96±0.08
7		74.50±4.01	69.89±6.10	79.01±3.65	74.03±13.41	81.49±3.49	85.16±5.83	97.87±1.52	96.11±3.74	98.61±1.05	95.97±2.50	56.61±36.41	88.78±31.35	98.53±1.66	97.84±1.37	99.96±**0.09**	98.99±0.44	99.85±0.22
8		79.29±4.09	85.12±2.90	79.24±2.85	81.91±5.93	92.02±1.74	88.77±3.88	92.87±3.46	97.02±2.35	97.41±1.05	98.83±0.79	82.70±24.22	77.01±40.15	97.75±0.81	99.05±**0.20**	99.68±0.27	97.36±0.89	98.92±0.72
9		99.72±0.10	99.39±0.44	75.95±8.76	99.74±**0.12**	98.84±1.36	91.99±2.73	98.27±0.94	98.97±1.25	97.23±0.78	96.61±1.68	43.09±26.39	87.61±20.54	97.66±1.57	89.81±5.30	99.78±0.14	97.73±1.53	98.31±0.73
AA (%) ↑		79.17±0.55	80.77±1.00	82.93±1.30	81.16±5.57	92.52±1.70	92.31±0.37	95.83±2.16	97.55±1.70	98.02±0.47	98.20±1.05	72.20±16.97	87.36±26.33	97.94±1.11	97.75±0.73	99.21±0.26	99.03±0.20	99.41±**0.21**
OA (%) ↑		83.41±0.41	85.53±0.37	88.80±0.57	84.37±0.71	94.88±0.33	94.82±0.84	96.93±0.73	98.58±0.33	98.79±0.37	99.03±0.21	71.35±18.54	88.96±19.20	97.46±0.80	99.12±0.50	99.59±0.14	99.20±0.11	99.70±**0.10**
k*100 (%) ↑		77.39±0.59	80.96±0.50	85.11±0.77	78.97±1.05	93.40±0.41	93.31±1.09	96.03±0.96	98.16±0.43	98.43±0.50	98.76±0.27	94.08±5.23	85.77±24.64	96.70±1.04	98.86±0.27	99.45±0.19	98.87±0.16	99.61±**0.13**
FWIoU (%) ↑		63.42±0.75	65.62±1.49	71.58±2.21	73.70±1.10	90.80±.55	90.63±1.41	94.38±1.24	94.33±.62	97.70±0.71	98.16±0.39	65.31±50.62	85.33±23.28	95.73±1.18	98.31±0.40	98.93±0.22	98.17±0.29	99.42±**0.19**
MIoU (%) ↑		66.00±0.61	67.62±1.49	72.04±2.11	71.68±1.15	87.75±0.90	87.91±2.17	92.65±1.44	95.96±1.01	96.63±1.01	96.65±0.79	58.87±22.54	82.34±26.08	92.38±1.57	96.29±1.19	98.23±0.37	97.68±0.20	99.00±**0.33**

TABLE IV

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, k, FWIOU, MIOU, AND THE ACCURACIES FOR EACH CLASS ON THE WHU-HI-HONGHU DATASET WITH 0.5% TRAINING SAMPLE. THE BEST RESULTS ARE COLORED SHADOW AND OPTIMAL STANDARD DEVIATION IN BOLD. RESULTS ON EXP1

Class No.	Color	Conventional Classifier			CNNs Based networks						GAN Based Networks		Transformer Based Networks		Unsupervised Network	Diffusion Model Based		DiffCRN
		KNN	RF	SVM	1D-CNN [104]	2D-CNN [78]	3D-CNN [80]	HybridSN [81]	SSRN [82]	SS-ConvNeXt [105]	MTGAN [39]	ADGAN [106]	SSFTT [86]	SSTN [85]	SC-SS-MTr [107]	DEMAE [64]	DKDMN [63]	
1		84.16±1.72	82.14±2.45	82.34±1.80	89.19±2.68	96.18±1.43	95.30±1.01	94.73±2.82	95.49±1.68	98.29±0.69	97.16±1.12	96.60±2.04	92.08±2.96	95.45±2.26	96.96±1.29	97.64±1.14	98.48±**0.67**	97.40±1.34
2		67.44±8.56	56.90±6.97	49.84±9.47	70.37±6.65	84.83±5.59	71.20±6.65	65.75±13.28	71.84±9.50	90.49±3.94	85.71±7.21	79.73±8.58	72.12±7.23	88.96±6.26	77.92±7.37	72.92±5.04	72.94±7.48	88.46±**2.34**
3		87.24±1.92	89.36±2.05	82.88±1.89	87.79±1.91	91.45±2.07	91.51±2.05	92.24±1.85	91.49±1.21	95.48±2.50	96.68±1.34	98.35±3.20	91.27±2.10	94.02±1.26	94.58±1.53	97.41±**1.06**	95.81±0.72	97.04±1.11
4		98.56±0.22	98.21±0.21	97.77±0.27	97.70±0.62	99.32±0.29	99.35±0.27	98.85±0.36	99.22±0.21	99.77±0.18	99.72±0.14	99.93±0.53	98.61±0.62	99.17±0.67	99.74±0.07	99.75±0.12	99.83±0.04	99.89±**0.07**
5		14.70±4.36	17.38±3.76	31.63±4.30	45.67±9.92	76.91±6.10	75.94±3.46	77.04±7.92	85.41±3.19	92.04±2.55	92.91±2.38	35.55±8.06	56.82±9.56	86.11±8.46	93.63±2.30	91.53±3.68	93.88±2.60	93.41±**1.82**
6		84.93±1.46	86.24±0.86	85.98±0.83	89.26±2.09	94.75±1.89	93.37±1.56	93.85±1.55	96.67±1.22	99.01±0.49	99.24±0.45	99.08±1.96	94.95±1.66	97.36±3.24	97.86±1.04	99.67±**0.13**	99.22±0.38	99.44±0.41
7		63.08±2.42	72.44±1.79	64.94±3.88	71.58±3.47	80.47±3.19	79.17±5.28	79.85±4.80	88.05±4.30	95.99±1.75	95.31±1.41	93.54±3.64	85.67±3.23	93.27±2.69	93.88±1.56	95.71±1.59	96.95±**0.57**	96.30±1.33
8		2.54±**0.73**	1.61±1.15	12.08±3.28	7.06±4.31	18.81±6.39	30.27±7.50	33.39±9.25	28.94±9.75	74.13±6.99	71.57±8.44	60.18±2.88	41.58±9.25	66.67±6.39	58.48±5.59	83.09±**4.95**	82.88±6.25	76.55±5.72
9		68.04±1.62	86.55±2.23	88.85±2.01	93.24±1.79	95.77±2.28	93.06±2.32	94.61±1.48	93.71±3.60	99.55±**0.42**	98.33±0.90	92.75±2.72	94.84±2.07	98.71±0.50	96.83±1.52	98.35±1.45	97.96±2.07	99.00±0.75
10		16.91±2.36	26.19±3.24	45.23±2.15	49.23±7.06	75.00±4.02	77.11±7.61	68.27±5.53	74.04±6.26	96.40±1.62	95.40±2.17	76.77±6.03	76.62±5.83	90.85±2.74	92.91±**1.48**	95.01±1.66	93.66±3.62	96.49±1.65
11		13.87±1.85	20.97±3.98	37.08±3.08	38.20±5.87	64.78±9.19	64.42±6.39	65.41±9.85	78.02±7.09	93.52±2.55	91.24±2.61	38.04±6.54	71.20±9.20	86.28±4.93	89.07±3.53	96.77±**1.01**	92.40±4.59	94.85±1.12
12		43.11±**2.53**	48.98±2.97	35.68±4.69	51.42±6.36	68.08±5.10	67.69±7.74	52.71±6.16	63.70±7.45	88.83±3.14	85.19±5.73	72.88±8.64	59.49±9.75	81.32±6.01	80.85±4.34	86.60±3.95	74.74±23.19	90.45±3.72
13		53.16±2.81	67.92±2.54	62.05±3.03	63.73±5.66	77.29±2.76	78.13±4.14	72.68±6.10	79.34±3.40	92.59±2.29	93.37±2.02	68.65±4.76	75.00±8.94	90.72±4.22	89.50±**1.52**	93.93±1.65	90.61±2.45	94.28±1.75
14		37.89±5.18	1.74±1.99	48.21±4.24	62.67±4.26	78.55±4.96	75.05±7.46	76.71±8.23	81.87±7.61	93.95±2.07	72.05±40.65	95.80±4.90	83.36±7.00	94.88±2.63	92.76±2.57	97.19±1.20	95.78±0.94	97.97±**0.89**
15		5.93±4.13	83.62±4.58	26.67±9.69	24.47±15.43	20.12±11.61	52.82±14.58	70.52±10.43	59.40±17.70	87.51±2.87	30.61±42.16	12.93±16.62	59.31±18.08	87.08±12.81	78.12±11.71	95.82±3.61	98.13±**1.33**	97.74±1.68
16		79.20±6.52	45.71±10.13	65.97±5.18	81.51±6.65	91.46±9.66	88.17±3.60	87.61±4.20	93.07±3.28	97.27±1.45	98.30±1.57	93.55±6.97	89.83±2.47	92.72±6.05	96.71±1.73	97.04±1.65	**98.27±1.04**	98.21±1.17
17		39.01±10.08	6.22±4.11	39.69±7.76	51.21±9.64	72.22±13.89	77.10±9.66	72.38±13.60	80.58±8.05	98.17±**1.67**	68.54±39.21	88.046±12.91	84.86±7.93	93.97±7.60	94.58±2.84	94.15±4.12	97.01±1.21	91.12±3.39
18		6.77±2.75	6.22±4.11	39.15±7.24	28.87±11.01	57.35±12.95	60.13±12.23	69.10±10.90	55.67±18.15	93.55±**2.26**	80.81±10.49	76.97±13.49	73.34±14.18	90.18±5.72	88.42±2.34	93.55±3.09	90.76±9.86	93.15±2.48
19		22.01±3.36	57.54±3.45	60.52±7.14	74.48±4.59	85.68±5.54	77.37±9.34	79.68±7.47	88.33±4.05	93.78±2.79	95.74±2.17	38.71±3.93	84.84±2.93	91.43±4.06	93.96±1.75	94.03±1.77	93.48±4.17	95.99±**0.87**
20		15.99±6.08	29.28±6.57	49.64±6.38	9.23±6.53	86.04±5.22	65.00±12.03	60.01±10.24	70.79±13.22	94.41±2.05	91.54±2.40	39.60±11.57	56.12±10.08	91.50±3.97	89.27±2.97	91.11±3.71	94.25±1.28	95.83±**1.13**
21		2.74±2.81	0.31±**0.37**	5.46±2.41	9.23±5.47	30.40±14.14	36.23±7.04	30.07±9.48	26.42±13.38	74.55±14.13	67.24±18.96	33.23±14.21	26.55±9.99	67.75±14.14	48.35±11.04	72.27±21.66	76.88±11.26	87.13±3.70
22		13.02±5.54	20.45±9.51	31.81±5.55	38.88±9.48	71.87±6.04	66.06±7.21	67.76±4.60	73.67±5.89	92.46±3.39	89.81±7.36	72.98±6.69	68.55±8.48	86.91±8.01	91.43±2.58	93.07±3.52	89.95±7.64	95.77±**1.94**
AA (%) ↑		41.83±0.68	47.60±1.00	51.98±1.34	58.62±2.35	73.56±2.29	73.38±1.66	72.87±1.46	75.94±2.84	92.81±0.86	86.20±2.15	71.09±2.12	74.41±2.59	89.33±1.25	87.99±0.97	92.16±0.42	91.99±1.98	94.39±**0.39**
OA (%) ↑		74.52±0.23	78.35±0.25	78.55±0.38	81.93±0.41	89.62±0.63	88.95±0.58	88.08±0.54	90.72±0.88	97.11±0.23	95.96±0.82	89.30±1.67	89.22±1.13	95.14±0.70	95.53±0.31	97.29±0.22	96.63±0.88	97.68±**0.16**
k*100 (%) ↑		66.75±0.33	72.15±0.36	72.56±0.51	77.05±0.58	86.91±0.79	86.06±0.73	84.97±0.69	88.28±1.12	96.36±0.29	94.90±1.04	86.72±1.76	86.40±1.41	93.89±0.88	94.35±0.39	96.57±0.30	95.74±1.11	97.08±**0.20**
FWIoU (%) ↑		20.69±0.61	26.59±0.89	30.93±1.17	72.27±0.68	83.03±0.91	82.08±0.74	80.80±0.76	84.56±1.26	94.59±0.43	92.74±1.23	82.90±1.43	82.04±1.66	91.31±1.09	91.96±0.60	94.90±0.29	93.76±1.43	95.60±**0.28**
MIoU (%) ↑		22.65±**0.67**	27.91±1.03	33.25±1.28	47.11±1.57	63.91±2.37	63.12±1.81	61.31±1.70	66.31±2.95	88.33±0.82	79.68±3.38	63.44±2.51	64.23±3.01	81.61±1.94	81.49±1.34	87.63±0.89	87.10±2.23	90.19±**0.67**

e.g., RF. The transformer-based network SSFTT cannot provide better performance than an unsupervised network with the pretraining method SC-SS-MTr. The proposed DiffCRN exhibits a performance that is superior to SC-SS-MTr, with an increase of approximately 1.7% in AA. Furthermore, under the circumstance of small categories, e.g., class 7 (grass-pasture-mowed) and class 9 (Oats), the SC-SS-MTr achieves poor performance, only 8.89% accuracy on class 9, although it is better in OA. Due to the advantage of the mask image modeling (MIM) method, DEMAE can perform better than DKDMN, which means that the joint effect of MIM and diffusion model can capture the spatial context information and data distribution. On the contrary, the proposed DiffCRN performs better on challenging classes (small-sample classes).

b) Performance on PU: Table III compares the experimental results to various classes on the PU dataset. From Table III, DiffCRN achieves the best accuracy (OA of 99.33%, AA of 99.26%, k of 99.31%, FWIoU of 98.81%, and MIoU of 98.60%) among the methods. The classification performance of SSRN, SSTN, and SC-SS-MTr is close on AA. Experimental results show that using a pretrained diffusion model to conduct a classification task successfully achieves high accuracy. The OA obtained by DiffCRN on the PU dataset outperforms the ones of SC-SS-MTr, SSFTT, MTGAN, SSRN, DEMAE, and DKDMN by 0.58%, 10.75%, 0.67%, 1.12%, 0.11%, and 0.5%, respectively.

c) Performance on WHU-Hi-HongHu: Table IV summarizes the comparative results on the WHU-Hi-HongHu dataset. We can see from Fig. 9 that the target distribution is more complex than the Indian Pines and PU datasets. Notably, the region is planted with different cultivars of the same crop type; for example, Chinese cabbage/cabbage (class 7/9) and Brassica chinensis/small Brassica chinensis (class 12/13). Not surprisingly, conventional classifier fails to complete this complex agricultural scene classification task with many classes of crops. We also see that an obvious noticeable rise in the direction of classic backbone (e.g., HybridSN and SSRN), GAN-based methods (e.g., MTGAN), transformer-based methods (e.g., SSTN), and SC-SS-MTr in terms of OA. The DDPM-based methods outperform the other classification models. Just as we talked about the cultivars of the same crop type, the accuracy obtained by DiffCRN on Chinese cabbage/cabbage outperforms the SSRN, MTGAN, SC-SS-MTr, and DEMAE by 8.25%/5.29%, 0.99%/0.67%, 2.42%/2.17%, and 0.59%/0.65%, respectively.

d) Performance on MUUFL: The quantitative results are listed in Table V. Both KNN, RF, and SVM outperform the GAN-based network ADGAN. The DiffCRN shows better accuracy than that of all other techniques, including GAN-, transformer-, and pretraining-based approaches, with OA, AA, k, FWIoU, and MIoU of 93.44 $\pm$ 0.28%, 82.77 $\pm$ 1.08%, and 91.56 $\pm$ 0.36%, 88.44 $\pm$ 0.44%, and 76.26 $\pm$ 1.06%, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, k, FWIOU, MIOU, AND THE ACCURACIES FOR EACH CLASS ON THE MUUFL DATASET WITH 3% TRAINING SAMPLE. THE BEST RESULTS ARE COLORED SHADOW AND OPTIMAL STANDARD DEVIATION IN BOLD. RESULTS ON EXP1

Class No.	Color	Conventional Classifier			CNNs Based networks						GAN Based Networks		Transformer Based Networks		Unsupervised Network	Diffusion Model Based		DiffCRN
		KNN	RF	SVM	1D-CNN [104]	2D-CNN [78]	3D-CNN [80]	HybridSN [81]	SSRN [82]	SS-ConvNeXt [105]	MTGAN [39]	ADGAN [106]	SSFTT [86]	SSTN [85]	SC-SS-MTr [107]	EMAE [64]	DKDMN [63]	
1		92.25±0.34	93.22±0.39	91.27±0.56	94.99±0.55	96.76±0.38	96.49±1.02	95.04±1.18	95.54±1.55	95.75±0.33	96.99±**0.30**	93.17±5.87	96.36±0.91	95.86±0.91	97.76±0.33	97.84±0.38	96.83±0.87	97.72±0.34
2		71.75±3.04	71.04±3.82	65.73±1.86	71.96±5.81	75.72±4.92	74.12±4.45	82.00±3.84	86.55±4.18	85.90±0.15	88.21±**1.40**	55.77±19.93	77.72±11.79	83.50±5.36	85.34±2.62	88.51±3.59	90.40±3.64	85.75±2.43
3		74.15±2.05	77.22±2.22	70.66±3.02	78.63±3.87	79.73±3.95	78.63±3.11	84.04±2.42	87.99±3.08	87.72±2.25	89.47±1.57	75.10±10.72	83.15±2.43	82.62±3.74	91.56±**1.09**	91.32±3.19	91.15±3.08	88.38±1.47
4		71.53±3.84	73.14±2.28	59.92±14.82	82.79±2.86	81.69±4.05	77.71±4.55	90.82±4.45	90.28±3.77	90.26±1.59	92.83±2.33	78.94±5.05	84.52±8.65	84.49±6.93	94.68±2.17	94.42±**1.64**	93.09±4.23	91.15±1.91
5		90.66±0.94	88.29±2.00	84.98±1.18	90.60±1.27	91.15±1.80	87.31±1.67	91.39±1.42	95.17±0.79	93.64±1.08	95.32±**0.46**	65.81±26.94	85.17±19.80	91.86±2.47	86.04±0.78	90.49±1.05	95.88±0.86	95.57±0.81
6		73.42±**1.42**	72.01±3.90	41.61±17.05	78.72±2.32	84.25±5.90	79.96±9.69	90.33±2.89	93.16±7.00	94.31±2.23	87.09±8.75	41.42±25.13	84.67±13.53	92.17±5.16	87.30±5.99	96.10±2.39	94.69±3.68	96.72±1.95
7		63.88±2.91	64.83±4.01	54.57±3.37	65.78±3.18	83.21±3.72	80.02±3.04	78.82±4.45	86.69±3.09	84.62±3.22	89.65±2.87	72.79±8.39	71.15±19.90	84.94±3.52	96.72±**1.53**	88.53±2.20	84.87±4.73	85.73±2.11
8		76.81±1.13	80.43±1.87	82.72±1.55	86.45±1.39	92.53±1.19	86.70±2.26	95.22±1.05	96.80±0.91	96.07±0.63	96.72±1.18	84.41±9.79	93.30±3.91	95.11±0.96	96.72±0.66	96.69±**0.50**	96.68±0.58	97.28±0.57
9		34.56±4.94	41.18±5.05	38.60±9.11	51.44±5.84	57.48±4.17	42.58±6.93	56.06±8.77	57.97±11.28	36.76±6.36	48.73±3.51	16.02±9.90	45.80±3.31	40.17±10.38	60.66±**4.15**	62.84±7.56	39.12±9.64	69.10±4.56
10		2.70±2.84	5.73±4.77	42.64±17.36	37.58±8.90	15.45±3.45	10.17±6.98	23.26±9.48	19.83±11.52	2.20±2.94	3.44±4.94	1.12±**2.51**	11.07±5.09	7.13±7.01	17.46±6.86	23.36±14.22	14.82±5.75	21.11±6.12
11		71.11±8.41	78.85±11.07	26.40±11.97	72.41±7.88	56.44±15.50	46.32±11.97	77.85±19.65	77.13±15.92	71.26±15.35	68.92±14.57	19.77±15.76	68.35±8.54	77.59±11.06	81.85±8.84	80.84±**4.40**	83.14±9.38	81.99±9.51
AA (%) ↑		65.71±**0.75**	67.81±1.36	59.92±5.03	73.76±1.53	74.13±4.46	69.09±1.11	78.62±1.94	80.65±5.74	76.23±1.98	77.94±2.04	54.97±7.36	72.84±6.68	75.95±1.89	80.65±1.46	82.81±1.32	80.06±1.87	82.77±1.08
OA (%) ↑		82.49±**0.20**	83.54±0.32	80.51±1.29	86.32±0.29	89.06±0.64	86.74±0.57	89.96±0.56	92.07±0.78	91.09±0.56	92.85±0.24	79.05±6.35	88.15±4.75	89.89±0.58	92.93±0.33	92.72±0.25	90.93±3.50	93.44±0.28
k*100 (%) ↑		76.83±**0.25**	78.85±0.41	74.30±1.68	82.01±0.40	85.92±0.85	82.86±0.69	87.15±0.71	89.85±1.00	88.42±0.77	90.81±0.30	76.53±9.05	84.43±7.07	86.88±0.78	90.96±0.44	91.02±0.33	90.64±1.22	91.56±0.36
FWIoU (%) ↑		44.37±1.66	48.50±2.00	37.98±5.56	77.08±0.51	81.64±1.00	78.18±0.66	83.17±0.81	86.40±1.23	84.54±0.93	87.39±0.38	71.19±8.91	80.09±7.49	83.04±0.94	87.61±0.55	87.18±**0.43**	87.14±1.54	88.44±0.44
MIoU (%) ↑		46.71±1.48	50.50±2.07	39.91±4.56	64.04±1.72	65.97±1.84	60.41±**0.96**	69.16±1.50	73.18±3.09	68.25±1.77	70.92±1.78	50.37±7.35	64.13±6.95	66.01±2.75	74.51±1.34	77.77±1.38	73.37±2.68	76.26±1.06

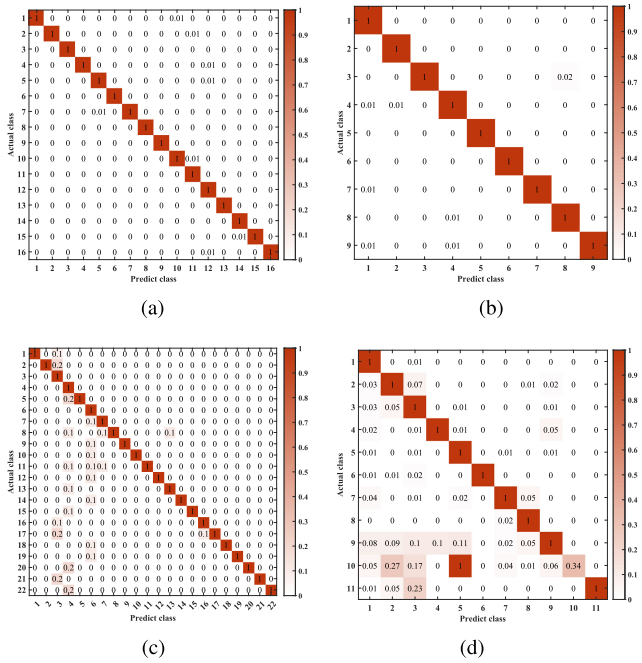


Fig. 10. Confusion matrix of the proposed DiffCRN on four datasets. N_{ij} of confusion matrix means that the ith class is classified to the jth class. Each row represents the class predicted; each column represents the true class. Results on Exp1. (a) Indian Pines. (b) PU. (c) WHU-Hi-HongHu. (d) MUUFL.

Furthermore, the classification confusion matrix of DiffCRN is shown in Fig. 10. From the diagonal elements, we can see that the number of each class varies slightly, except class 10 (road) in the MUUFL dataset.

2) Visual Evaluation: Figs. 11–14 show the image domain classification results of different approaches on the four datasets. Our goal is to perform a qualitative evaluation of the compared methods. Overall, the visual effects produced by DiffCRN are better than other methods. Conventional classifiers, such as KNN, RF, SVM, and 1-D CNN, provide classification maps with salt and pepper noise around the boundary areas because they only exploit spectral information. Classic backbone networks, such as HybridSN and SSRN, improve this phenomenon and achieve better visual performance. MTGAN and ADGAN provide smoother boundaries due to the encoder–decoder architecture, resulting in losing spatial information. Transformer-based methods, e.g., SSTN, can extract more abstract information in sequential representation, so they provide better classification maps than classic backbone networks. The classification map generated by SC-SS-MTr provides discrete raster grid points, maybe due to the masked image modeling mechanism of SC-SS-MTr in the pretraining stage, resulting in some error reconstruction of the raster, while the DEMAE with masked image modeling can better deal with the problem. With reference to the RGB composite image, DiffCRN characterizes texture and edge details better than other techniques.

Furthermore, 2-D graphical visualization maps depict the features extracted by different methods and the proposed DiffCRN. Fig. 15(a)–(f) show MTGAN, SSRN, SSFTT, SC-SS-MTr, DEMAE, and DiffCRN on four datasets, respectively. Using the t-distributed stochastic neighbor embedding (t-SNE), we can clearly see that the test data of similar categories gather together, and intraclass variance is minimized on our proposed DiffCRN.

D. Performance Over Limited Training Sample Size

To further show the classification performance, we conduct Exp2 on four datasets. Twenty samples are selected for each class, as listed in Table I. The results are shown in Table VI. As a matter of course, even with a limited number of training samples, the proposed DiffCRN exhibits significantly better classification performance than the other methods, especially a significant enhancement in terms of MIoU. Notably, the classification performance of the Indian Pines dataset is largely improved compared with the previous SOTA pretrained method based on masked image modeling, e.g., SC-SS-MTr, in terms of OA (92.03% versus 90.04%), AA (96.25% versus 94.41%), k (91.26% versus 88.69%), FWIoU (86.23% versus 83.04%), and MIoU (85.99% versus 78.66%).

E. Model Analysis

1) Ablation Study: In this section, we analyze the effect of the key components and the objective function in our method.

a) With or without spatial–spectral diffusion model: To demonstrate the effectiveness of DiffCRN, we designed the following three comparative experiments.

Case 1: We only feed the raw all-band hyperspectral instance to the classification backbone without spectral–spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

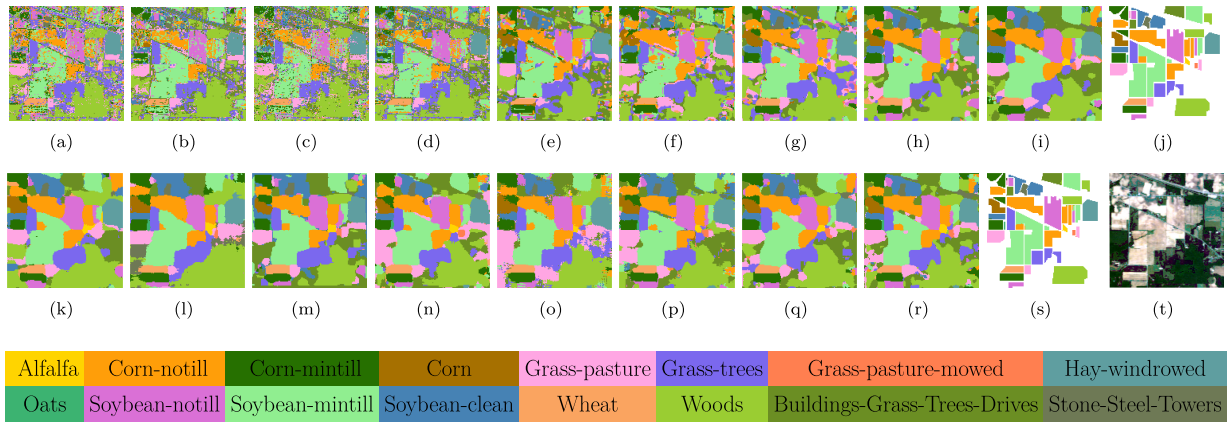


Fig. 11. Classification maps of the Indian Pines dataset. (a) KNN (OA = 68.20%). (b) RF (OA = 74.85%). (c) SVM (OA = 74.63%). (d) 1-D CNN (OA = 73.75%). (e) 2-D CNN (OA = 88.83%). (f) 3-D CNN (OA = 87.90%). (g) HybridSN (OA = 96.18%). (h) SSRN (OA = 97.81%). (i) SS-ConvNeXt (OA = 98.20%). (j) Ground-truth map. (k) MTGAN (OA = 97.20%). (l) ADGAN (OA = 97.63%). (m) SSFTT (OA = 95.12%). (n) SSTN (OA = 97.15%). (o) SC-SS-MTr (OA = 97.33%). (p) DEMAE (OA = 98.43%). (q) DKDMN (OA=). (r) DiffCRN (OA = 99.33%). (s) Label-domain classification map of DiffCRN. (t) RGB composite image. Results on Exp1.

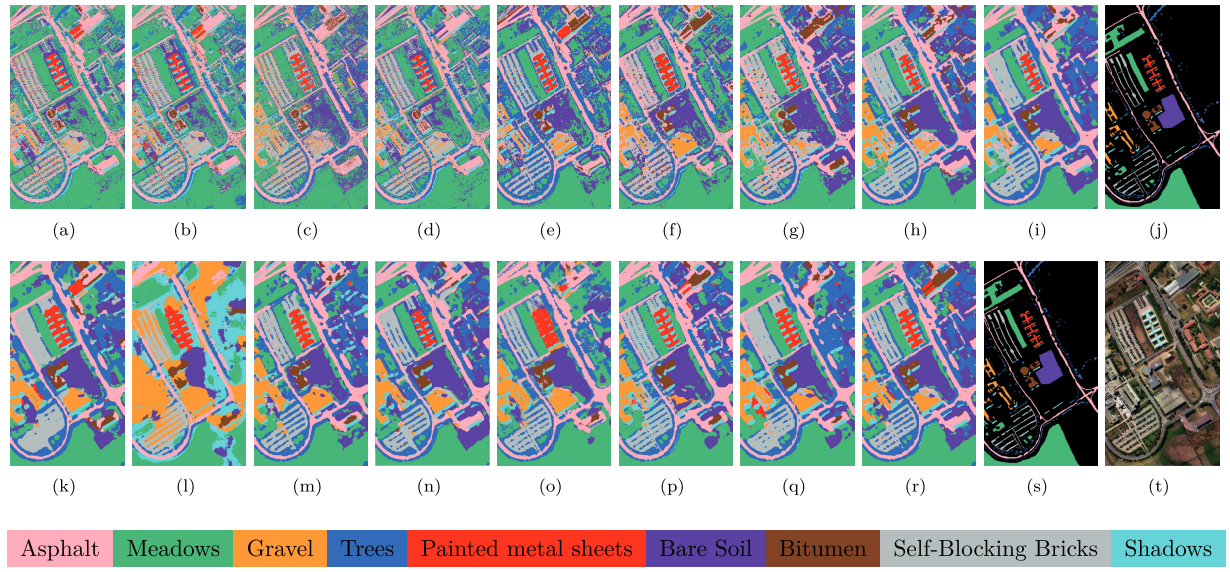


Fig. 12. Classification maps of the PU dataset. (a) KNN (OA = 83.41%). (b) RF (OA = 85.53%). (c) SVM (OA = 88.80%). (d) 1-D CNN (OA = 84.37%). (e) 2-D CNN (OA = 94.88%). (f) 3-D CNN (OA = 94.82%). (g) HybridSN (OA = 96.93%). (h) SSRN (OA = 98.58%). (i) SS-ConvNeXt (OA = 98.79%). (j) Ground-truth map. (k) MTGAN (OA = 99.03%). (l) ADGAN (OA = 71.35%). (m) SSFTT (OA = 88.96%). (n) SSTN (OA = 97.46%). (o) SC-SS-MTr (OA = 99.12%). (p) DEMAE (OA = 99.59%). (q) DKDMA (OA = 99.20%). (r) DiffCRN (OA = 99.70%). (s) Label-domain classification map of DiffCRN. (t) RGB composite image. Results on Exp1.

diffusion contrastive representation learning, AWAM, and CTSSFM, noted as "Baseline" in Table VII.

Case 2: We feed the raw all-band hyperspectral instance to the proposed classifier. What needs illustration is that the clean all-band hyperspectral instances are processed by the AWAM and CTSSFM, noted as "Raw all-band" in Table VII.

Case 3: We feed the features extracted by the pretrained DiffCRN to the proposed classifier, noted as "Diffusion features" in Table VII.

As shown in Table VII, the classification results on four datasets are presented. The results demonstrate that using diffusion features as input significantly outperforms the use of raw features and baseline methods, for example, the improvement in terms of OA, with an increase of approximately 10.20% on the Indian Pines dataset, 9.02% on the PU dataset, 8.97% on WHU-Hi-HongHu, and 4.45% on MUUFL. Furthermore, the more obvious improvement is in MIoU, with an increase of approximately 17.51%, 17.68%, 28.56%, and 13.76% on these datasets, which indicates that the extracted features have stronger discrimination, resulting in lower misclassification ability and the effectiveness of using spectral–spatial diffusion contrastive representation learning.

b) With or without AWAM and CTSSFM: In DiffCRN, we use AWAM and CTSSFM to fuse information adaptively from identical and cross-diffusion timesteps t, achieving a more detailed classification map and obtaining high-accuracy category prediction.

To verify the effectiveness of AWAM and CTSSFM, an ablation experiment is designed to ascertain the effectiveness. In detail, four comparisons are listed as follows:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

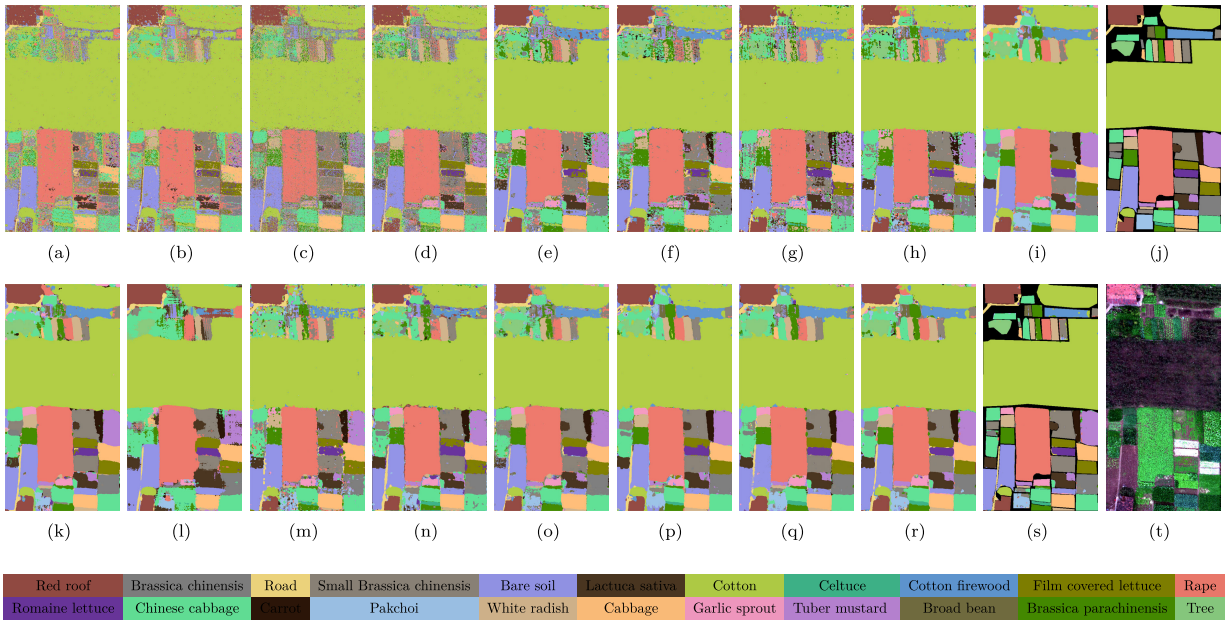


Fig. 13. Classification maps of the WHU-Hi-HongHu dataset. (a) KNN (OA = 74.52%). (b) RF (OA = 78.35%). (c) SVM (OA = 78.55%). (d) 1-D CNN (OA = 81.93%). (e) 2-D CNN (OA = 89.62%). (f) 3-D CNN (OA = 88.95%). (g) HybridSN (OA = 88.08%). (h) SSRN (OA = 90.72%). (i) SS-ConvNeXt (OA = 97.11%). (j) ground-truth map. (k) MTGAN (OA = 95.96%). (l) ADGAN (OA = 89.30%). (m) SSFTT (OA = 89.22%). (n) SSTN (OA = 95.14%). (o) SC-SS-MTr (OA = 95.53%). (p) DEMAE (OA = 97.29%). (q) DKDMA (OA = 96.63%). (r) DiffCRN (OA = 97.68%). (s) Label-domain classification map of DiffCRN. (t) RGB composite image. Results on Exp1.

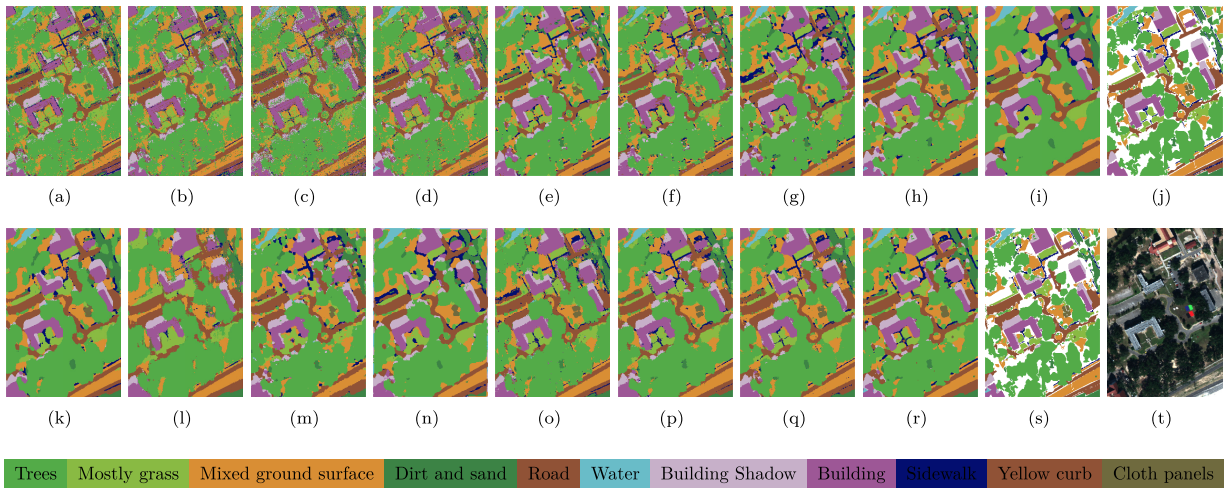


Fig. 14. Classification maps of the MUUFL dataset. (a) KNN (OA = 82.49%). (b) RF (OA = 83.54%). (c) SVM (OA = 80.51%). (d) 1-D CNN (OA = 86.32%). (e) 2-D CNN (OA = 89.06%). (f) 3-D CNN (OA = 86.74%). (g) HybridSN (OA = 89.46%). (h) SSRN (OA = 92.07%). (i) Ground-truth map. (j) SS-ConvNeXt (OA = 91.09%). (k) MTGAN (OA = 92.85%). (l) ADGAN (OA = 79.05%). (m) SSFTT (OA = 88.15%). (n) SSTN (OA = 89.89%). (o) SC-SS-MTr (OA = 92.93%). (p) DiffCRN (OA = 93.44%). (q) Label-domain classification map of DiffCRN. (r) RGB composite image. Results on Exp1.

1) only linear classifier; 2) AWAM with linear classifier; 3) CTSSFM with linear classifier; and 4) AWAM and CTSSFM with linear classifier. Table VIII shows the relevant experimental results on the Indian Pines dataset. Experimental results show that w/o AWAM&CTSSFM performs worse, possibly due to the loss of the multitimestep representations fusion. What is more, the use of AWAM or CTSSFM can significantly improve the model performance. Furthermore, the combined utilization of AWAM and CTSSFM can be further improved. More specifically, adding the AWAM and CTSSFM improves the OA, AA, k, FWIou, and MIoU scores on the Indian Pines by 2.39%, 3.37%, 2.48%, 4.18%, and 5.23%.

c) Compound loss function or single objective function of spatial–spectral diffusion model: To thoroughly evaluate the performance of the proposed method with the compound loss, we compare five cases of the objective function of the spatial–spectral diffusion model and evaluate the classification performance on the Indian Pine dataset. These will be referred to as the following cases.

Case 1: The mse is used as the objective function of the spatial–spectral diffusion model, which is also the original and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

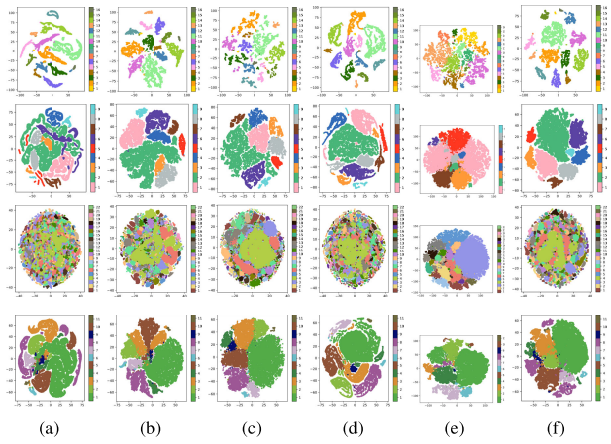


Fig. 15. 2-D graphical visualization of the features extracted by different methods and proposed DiffCRN through t-SNE on the four datasets. (Top to bottom) Indian Pines, PU, WHU-Hi-HongHu, and MUUFL. (a) MTGAN. (b) SSRN. (c) SSFTT. (d) SC-SS-MTr. (e) DEMAE. (f) DiffCRN. Results on Exp1.

TABLE VI

QUANTITATIVE PERFORMANCE OF DIFFERENT CLASSIFICATION METHODS IN TERMS OF OA, AA, k, FWIOU, AND MIOU ON THE FOUR DATASETS. THE BEST RESULTS ARE IN BOLD AND COLORED SHADOW, AND SUBOPTIMALS ARE IN UNDERLINED. RESULTS ON EXP2

Dataset / Metrics	Indian Pines				
	MTGAN [39]	SSRN [82]	SSFTT [86]	SC-SS-MTr [107]	**DiffCRN**
OA (%) ↑	88.60±2.60	89.59±1.23	83.99±3.03	90.04±1.96	**92.03±1.12**
AA (%) ↑	92.90±1.19	94.78±2.03	84.53±3.85	94.41±0.91	**96.25±0.66**
k*100 (%) ↑	87.46±2.78	88.57±1.33	82.35±3.26	88.69±2.20	**91.26±1.23**
FWIoU (%) ↑	80.92±3.68	82.33±1.75	74.10±3.82	83.04±3.25	**86.23±1.76**
MIoU (%) ↑	77.39±3.37	81.39±2.24	74.41±2.48	78.66±3.64	**85.99±2.91**
Dataset / Metrics	Pavia University				
	MTGAN	SSRN	SSFTT	SC-SS-MTr	**DiffCRN**
OA (%) ↑	**96.66±0.94**	91.88±1.24	87.81±2.16	95.86±2.33	95.97±0.82
AA (%) ↑	96.02±1.26	93.60±3.41	92.25±2.70	96.72±1.33	**97.06±0.36**
k*100 (%) ↑	**95.60±1.24**	89.42±1.64	84.44±2.68	94.59±3.01	94.73±1.04
FWIoU (%) ↑	**93.82±1.60**	85.74±2.12	79.02±3.15	92.66±3.99	92.53±1.40
MIoU (%) ↑	91.15±2.41	85.42±1.30	79.04±3.64	91.19±4.11	**92.33±0.81**
Dataset / Metrics	WHU-Hi-HongHu				
	MTGAN	SSRN	SSFTT	SC-SS-MTr	**DiffCRN**
OA (%) ↑	92.11±1.87	86.48±1.61	75.94±3.93	88.14±2.03	**93.19±0.56**
AA (%) ↑	91.23±1.79	85.88±2.34	76.02±1.69	90.02±1.23	**92.67±0.55**
k*100 (%) ↑	90.10±1.10	83.28±1.92	70.95±4.23	87.36±1.40	**91.47±0.67**
FWIoU (%) ↑	86.94±1.32	76.54±2.25	66.97±4.19	83.90±3.53	**88.64±0.76**
MIoU (%) ↑	75.57±3.10	66.97±2.52	50.13±2.76	70.36±3.47	**78.57±1.02**
Dataset / Metrics	MUUFL				
	MTGAN	SSRN	SSFTT	SC-SS-MTr	**DiffCRN**
OA (%) ↑	81.55±1.52	81.76±2.97	74.51±1.51	81.14±2.63	**83.75±1.53**
AA (%) ↑	76.14±3.84	82.38±5.20	75.82±4.48	81.80±1.92	**84.12±1.28**
k*100 (%) ↑	76.05±1.93	76.80±3.45	67.82±1.76	76.03±3.29	**79.21±1.79**
FWIoU (%) ↑	71.68±2.13	72.71±3.70	63.80±1.86	72.23±3.14	**75.46±1.91**
MIoU (%) ↑	54.55±3.24	59.04±2.76	47.04±1.51	54.40±3.98	**59.19±1.67**

TABLE VII

CLASSIFICATION PERFORMANCE WITH DIFFERENT INPUTS ON FOUR DATASETS. THE BEST RESULTS ARE IN BOLD AND COLORED SHADOW

Metrics / Case	Indian Pines				
	OA	AA	Kappa	FWIoU	MIoU
Baseline	89.13±2.06	86.16±2.68	88.68±2.21	82.31±3.14	81.09±2.78
Raw all-band	96.07±0.99	94.54±2.43	95.85±1.11	93.08±1.77	92.39±2.65
Diffusion feature	**99.33±0.16**	**99.26±0.37**	**99.31±0.16**	**98.81±0.28**	**98.60±0.55**
Metrics / Case	Pavia University				
	OA	AA	Kappa	FWIoU	MIoU
Baseline	90.68±1.58	91.15±1.78	88.01±2.07	83.70±2.49	81.32±3.03
Raw all-band	94.76±0.58	95.36±1.10	93.21±0.74	90.71±0.99	86.86±1.33
Diffusion feature	**99.70±0.10**	**99.41±0.21**	**99.61±0.13**	**99.42±0.19**	**99.00±0.33**
Metrics / Case	WHU-Hi-HongHu				
	OA	AA	Kappa	FWIoU	MIoU
Baseline	88.71±0.65	72.20±2.12	85.78±0.83	81.05±0.94	61.63±2.02
Raw all-band	95.56±0.57	88.53±1.52	94.40±0.72	91.95±0.91	82.42±1.91
Diffusion feature	**97.68±0.16**	**94.39±0.39**	**97.08±0.20**	**95.60±0.28**	**90.19±0.68**
Metrics / Case	MUUFL				
	OA	AA	Kappa	FWIoU	MIoU
Baseline	88.99±0.75	73.02±2.39	85.85±0.95	81.76±1.07	62.50±2.71
Raw all-band	90.38±0.63	79.61±2.47	88.19±0.83	84.50±0.94	69.53±2.33
Diffusion feature	**93.44±0.28**	**82.77±1.08**	**91.56±0.36**	**88.44±0.44**	**76.26±1.06**

TABLE VIII

ABLATION STUDY FOR THE AWAM AND CTSSAM ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE IN BOLD AND COLORED SHADOW

Metric	OA	AA	Kappa	FWIoU	MIoU
w/o AWAM&CTSSFM	96.94±0.36	95.89±0.77	96.83±0.38	94.63±0.60	93.37±1.71
w/o CTSSAM	97.47±0.38	96.37±1.07	97.40±0.39	95.58±0.62	93.86±1.64
w/o AWAM	99.15±0.25	99.0±1.23	99.13±0.26	98.50±0.45	98.22±1.13
w/ AWAM&CTSSAM	**99.33±0.16**	**99.26±0.37**	**99.31±0.16**	**98.81±0.28**	**98.60±0.55**

w/: with, w/o: without

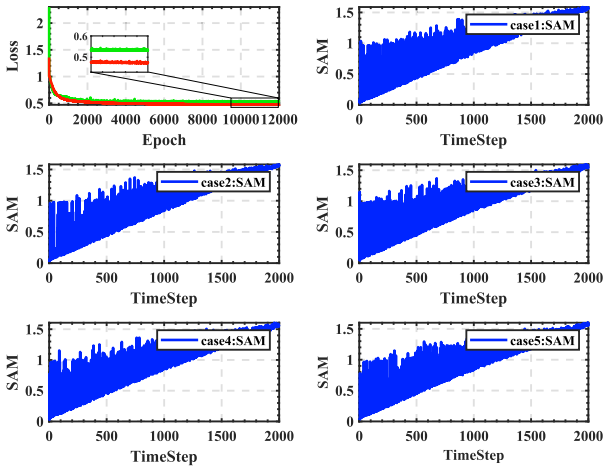


Fig. 16. (Top left corner) Comparison between mse loss and LAE loss on the Indian Pines dataset. Red line: LAE; blue line: mse. Rest: learned SAM value under different cases on the Indian Pines dataset. The smaller the SAM, the higher the similarity.

unmodified loss in DDPM [43] shown in (18). For the sake of simplicity, $\mathcal{L} = \mathcal{L}_{\text{MSE}}$.

Case 2: The modified mse loss function, named LAE, is used as the objective function, as shown in (19). For the sake of simplicity, $\mathcal{L} = \mathcal{L}_{\text{diff}}$.

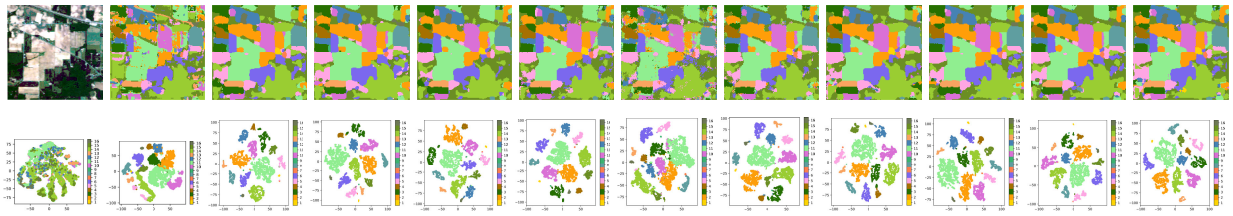
Case 3: Considering CL loss, the loss function can be written as $\mathcal{L} = \mathcal{L}_{\text{diff}} + \mathcal{L}_{\text{con}}$.

Case 4: Considering the uncertainty of optimization objective, and using CL loss, the loss function can be written as $\mathcal{L} = e^{-s_{\text{diff}}}\mathcal{L}_{\text{diff}} + \mathcal{L}_{\text{con}} + s_{\text{diff}}$.

Case 5: Considering the uncertainty of optimization objective and reconstruction loss in (20), and using CL loss, the loss function can be writen as $\mathcal{L} = e^{-s_{\text{diff}}}\mathcal{L}_{\text{diff}} + e^{-s_{\text{rec}}}\mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{con}} + s_{\text{diff}} + s_{\text{rec}}$.

The results are listed in Table IX. Comparing case 2 with case 1, there has been a significant improvement in all metrics, e.g., 1% improvement in OA. Thus, the modification of the raw objective function of the diffusion model, e.g., (19), can better reconstruct the added Gaussian noise and model the local–global spectral–spatial relationship of HSI by putting more emphasis on heavily penalizing the big gap between $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}_\theta(\mathcal{H}_t, t)$, as shown in top left corner of Fig. 16. Cases 2–4 have similar results. In case 5, by comprehensively considering the uncertainty of optimization objective and reconstruction loss in (20), and using CL loss, the metrics are still improving, especially in terms of AA, an increase of about 0.6%. It means that more discriminative features are obtained by optimizing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.



(a) Raw (b) Baseline (c) S1 (d) S2 (e) S3 (f) S4 (g) S5 (h) S6 (i) S7 (j) S8 (k) S9 (l) S10

Fig. 17. Classification map under different sets of timesteps t as listed on Table X on the Indian Pines dataset. Each column corresponds to Table X in order. The second and fourth row is the corresponding 2-D graphical visualization of the features extracted under different sets of timesteps through t-SNE. (a) Distribution characteristics of raw labeled data. (b) OA = 89.13%. (c) OA = 99.27%. (d) OA = 99.18%. (e) OA = 98.68%. (f) OA = 98.80%. (g) OA = 94.09%. (h) OA = 98.69%. (i) OA = 99.33%. (j) OA = 99.23%. (k) OA = 99.18%. (l) OA = 98.45%. Zoom in for the best view.

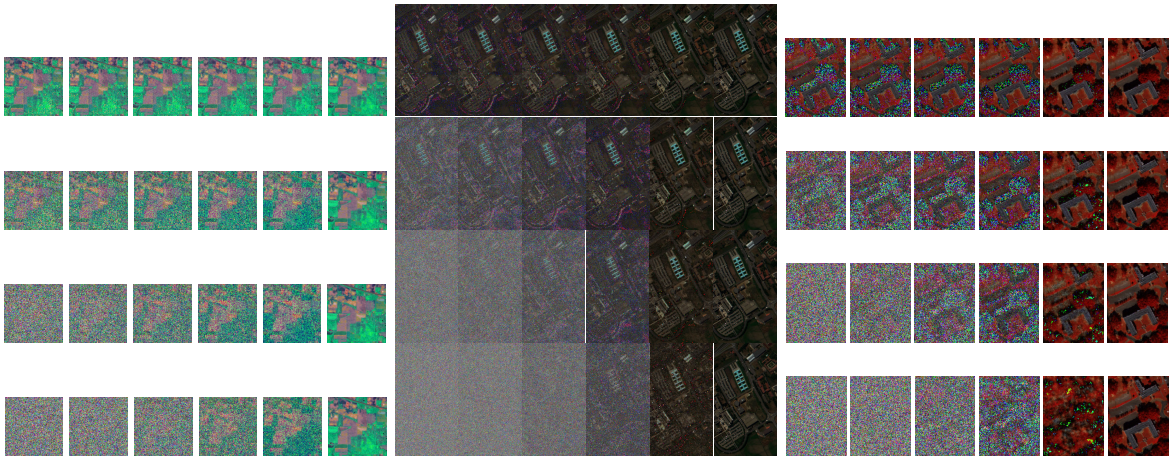


Fig. 18. False-color images of the reconstructed Indian Pines, subset of PU and MUUFL datasets. (Top to bottom): t = 110, 510, 1010, and 1610, corresponding to different timesteps by reverse spectral–spatial diffusion process. The first column is $\mathcal{H}_t$. The penultimate column is the reconstructed $\mathcal{H}_0$. The second, third, and fourth columns are the intermediate results sampled by (10). The last column is the original hyperspectral false-color image $\mathcal{H}_0$. (a) Raw. (b) Baseline. (c) S1. (d) S2. (e) S3. (f) S4. (g) S5. (h) S6. (i) S7. (j) S8. (k) S9. (l) S10.

TABLE IX

CLASSIFICATION PERFORMANCE WITH DIFFERENT SETS OF OBJECTIVE FUNCTION t ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE IN BOLD AND COLORED SHADOW

Objective function	timestep	OA	AA	Kappa	FWIoU	MIoU
Case 1	[0, 1, 2, 4, 5]	98.13±0.22	97.69±0.69	98.08±0.23	96.72±0.38	95.57±0.63
Case 2	[0, 1, 3, 2, 4]	99.25±0.16	98.65±0.54	99.23±0.16	98.68±0.28	97.91±0.40
Case 3	[0, 1, 4, 7, 13]	99.15±0.16	98.91±0.41	99.12±0.16	98.48±0.28	98.02±0.77
Case 4	[1, 2, 5, 6, 10]	99.20±0.22	98.82±0.82	99.18±0.22	98.58±0.38	97.94±0.87
Case 5	[0, 1, 2, 3, 5]	**93.33±0.16**	**99.25±0.40**	**99.31±0.16**	**98.81±0.28**	**98.60±0.55**

[1] Choosing timestep t according to the top 5 results of SAM value.

TABLE X

CLASSIFICATION PERFORMANCE WITH DIFFERENT SETS OF TIMESTEPS t ON THE INDIAN PINES DATASET WITH 10% TRAINING SAMPLES. THE BEST RESULTS ARE IN BOLD AND COLORED SHADOW

Index	timestep [1]	Metric				
		OA	AA	Kappa	FWIoU	MIoU
S1	[0, 1, 2, 3, 4]	99.27±0.17	99.08±0.40	99.25±0.18	98.70±0.31	98.43±0.62
S2	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]	99.18±0.26	99.03±0.5	99.16±0.26	98.54±0.45	98.29±0.64
S3	[0, 1, 2, 3, 4, ... , 26, 27, 28, 29]	98.68±0.28	98.29±0.60	98.66±0.29	97.69±0.48	96.99±1.33
S4	[50, 100, 200, 300, 500]	98.80±0.24	98.91±0.51	98.77±0.25	97.87±0.42	97.85±0.75
S5	[700, 900, 1100, 1300, 1500]	94.09±0.64	93.27±2.64	93.78±0.70	89.78±1.11	89.34±2.67
S6	[50, 200, 500, 900, 1500]	98.69±0.19	98.57±0.75	98.66±0.19	97.68±0.32	97.37±0.97
S7	Top5 of SAM value	**99.33±0.16**	**99.26±0.37**	**99.31±0.16**	**98.81±0.28**	**98.60±0.55**
S8	Top10 of SAM value	99.23±0.11	98.99±0.39	99.21±0.11	98.62±0.19	98.26±0.48
S9	Top30 of SAM value	99.18±0.11	98.83±0.74	99.16±0.11	98.55±0.19	98.22±0.80
S10	Top100 of SAM value	98.45±0.67	98.36±1.88	98.42±1.16	97.28±1.16	97.21±0.74

[1] [50, 100, 200, 300, 500], [700, 900, 1100, 1300, 1500] and [50, 200, 500, 900, 1500] means manually select timestep t, while Top5, Top10, Top30 of SAM value mean adaptively select timestep t according to the top few value of SAM.

the compound loss function. The diffusion timestep t is determined by the learned SAM, as shown in Fig. 16.

d) Select t adaptively or manually: We analyze the representations produced by the noise predictor $\boldsymbol{\epsilon}_\theta(\mathcal{H}_t, t)$ for five different sets of diffusion timestep t. The first six rows of the Table X are manually selected timestep sets, while the rest are determined by the learned SAM. As shown in Table X, we can clearly see the performance difference under different sets of timesteps t. Specifically, it can be seen that the following conditions hold.

1) When the timestep t is manually selected, larger values of t tend to extract features with significantly more noise, resulting in poorer performance compared to smaller timesteps. For example, timestep with [700, 900, 1100, 1300, 1500], only 94.09% accuracy in term of OA, which is nearly 5% lower than that of timestep with [50, 100, 200, 300, 500] and timestep with small (i.e., [50, 100, 200, 300, 500]) and medium t (i.e., [700, 900, 1100, 1300, 1500]) have approximate performance.
2) Timestep t determined by the learned SAM value performs better than manually assigned t and further performs better than timestep with [0, 1, 2, 3, 4], [0, 1, 2, 3, 4, 5, 6, 7, 8, 9] and [0, 1, 2, 3, 4, . . . , 26, 27, 28, 29], which demonstrates the effectiveness of building the intermediate features according to SAM. Three different scenarios according to the top few values of SAM have similar performance.
3) Furthermore, we visualize the features in 2-D space using t-SNE to show the diversity of features at different sets of timesteps t. Although the original distribution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
COMPUTATIONAL COST AND MODEL SIZE ACROSS DIFFERENT DIFFUSION ARCHITECTURES

	Indian Pines			
	Spectraldiff	DKDMN	DEMAE	DiffCRN
FLOPs	1.557T	97.308G	68.595M	294.601M
Params	4.300M	4.300M	783.288K	4.672M
	Pavia University			
	Spectraldiff	DKDMN	DEMAE	DiffCRN
FLOPs	809.604G	50.600G	68.595M	467.173M
Params	4.300M	4.300M	783.288K	4.564M
	WHU-Hi-HongHu			
	Spectraldiff	DKDMN	DEMAE	DiffCRN
FLOPs	4.110T	256.893G	68.595M	741.513M
Params	4.300M	4.300M	783.288K	4.751M
	MUUFL			
	Spectraldiff	DKDMN	DEMAE	DiffCRN
FLOPs	996.432G	62.277G	68.595M	460.091M
Params	4.520M	4.300M	783.288K	4.520M

characteristics of raw data are complex, i.e., Fig. 17(a), we can observe that by using SAM to extract features, samples of similar categories gather together, and intraclass variance is minimized, i.e, Fig. 17(i)–(l). On the contrary, it can be seen that the features learned by manually assigned t are less discriminative, i.e., Fig. 17(g), as well as the baseline method, i.e., Fig. 17(b), which also reflects in the corresponding classification map full of salt and pepper noise.

2) Diffusion Model Analysis: Since the diffusion model can convert the Gaussian noise distribution into the target distribution $\mathcal{H}_0$, in this section, we analyze the restoration and reconstruction performance of DiffCRN; in other words, we use the pretrained diffusion model to recover and reconstruct the hyperspectral imagery data. As shown in Fig. 18, the first column is $\mathcal{H}_t$, which is almost completely destroyed as the diffusion steps increase, e.g., $t = 1010, 1610, 1790$. Even so, from a visual perspective, the diffusion model can basically recover the details of hyperspectral imagery.

3) Model Parameter Analysis: To compare with the computational cost and model size, we set the batch size equal to 2 for three diffusion-based methods, e.g., Spectraldiff, DKDMN, DEMAE, and our proposed method, DiffCRN, to calculate the number of parameters and the floating-point operations (FLOPs). As we can see from Table XI, spectraldiff, 3DCONV-based model, has a huge amount of FLOPs; for example, with the shape of [2, 1, 270, 64, 64] as the model input, the FLOPs will reach 4.110T, leading to slow model training and inference. During model training, spectraldiff will cost around seven days to pretrain, which is much longer than DiffCRN. DKDMN is also implemented based on 3DCONV, so the computational cost is also higher than DiffCRN. In addition, due to the PCA reduction, DEMAE reduces all four datasets into 36 dimensions, which hugely reduces the model parameters and FLOPs.

V. CONCLUSION

In this study, we leverage the DDPM to model the local–global spectral–spatial relationships in an unsupervised manner. To narrow the distance between added Gaussian noise and reconstructed ones, instead of the widely used mse loss function in DDPM, we use a logarithm loss function (LAE) to optimize the diffusion model, achieving better performance. To improve the instance-level and interclass discriminability of the diffusion model, we introduce CL to build discriminative representations, which has a positive effect on the classification task. Furthermore, instead of selecting manually the timestep t to build multitimestep representations, we propose a simple but effective way to adaptively choose multitimestep representations based on pixel-level SAM. Simultaneously, the proposed AWAM and CTSSFM modules can be effectively combined with DiffCRN to get refined features. Extensive evaluations show that the proposed model, DiffCRN, performs favorably against state-of-the-art supervised and unsupervised methods on HSIC on the four widely used datasets.

However, the proposed methods utilized a two-stage learning strategy, which leads to a heavy training burden and inconvenience when used for practice. In future work, we will improve the proposed methods to be an end-to-end framework and explore a conditional diffusion model to generate realistic HSIs via language description.

REFERENCES

[1] B. Lu, P. Dao, J. Liu, Y. He, and J. Shang, "Recent advances of hyperspectral imaging technology and applications in agriculture," *Remote Sens.*, vol. 12, no. 16, p. 2659, Aug. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/16/2659

[2] C. Wang et al., "A review of deep learning used in the hyperspectral image analysis for agriculture," *Artif. Intell. Rev.*, vol. 54, no. 7, pp. 5205–5253, Oct. 2021, doi: 10.1007/s10462-021-10018-y.

[3] X. Tong, H. Xie, and Q. Weng, "Urban land cover classification with airborne hyperspectral data: What features to use?" *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 10, pp. 3998–4009, Oct. 2014.

[4] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[5] R. J. Murphy, S. Schneider, and S. T. Monteiro, "Consistency of measurements of wavelength position from hyperspectral imagery: Use of the ferric iron crystal field absorption at ~900 nm as an indicator of mineralogy," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2843–2857, May 2014.

[6] C. A. Bishop, J. G. Liu, and P. J. Mason, "Hyperspectral remote sensing for mineral exploration in Pulang, Yunnan province, China," *Int. J. Remote Sens.*, vol. 32, no. 9, pp. 2409–2426, May 2011, doi: 10.1080/01431161003698336.

[7] S. Shi, M. Zhao, L. Zhang, Y. Altmann, and J. Chen, "Probabilistic generative model for hyperspectral unmixing accounting for endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516915.

[8] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[9] C. Yu, C. Liu, M. Song, and C.-I. Chang, "Unsupervised domain adaptation with content-wise alignment for hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[10] C. Persello and L. Bruzzone, "Active and semisupervised learning for the classification of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6937–6956, Nov. 2014.

[11] D. Tuia, M. Volpi, L. Copa, M. Kanevski, and J. Munoz-Mari, "A survey of active learning algorithms for supervised remote sensing image classification," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 606–617, Jun. 2011.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

[12] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[13] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[14] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[15] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[16] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[17] X. He and Y. Chen, "Optimized input for CNN-based hyperspectral image classification using spatial transformer network," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1884–1888, Dec. 2019.

[18] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[19] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[20] D. Ibanez, R. Fernandez-Beltran, F. Pla, and N. Yokoya, "Masked auto-encoding spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542614.

[21] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[22] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[23] M. E. Paoletti et al., "Capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2145–2160, Apr. 2019.

[24] M. E. Paoletti, S. Moreno-Álvarez, and J. M. Haut, "Multiple attention-guided capsule networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520420.

[25] B. Liu, A. Yu, X. Yu, R. Wang, K. Gao, and W. Guo, "Deep multiview learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7758–7772, Sep. 2021.

[26] X. Li et al., "Multi-view learning for hyperspectral image classification: An overview," *Neurocomputing*, vol. 500, pp. 499–517, Aug. 2022.

[27] U. Patel, H. Dave, and V. Patel, "Hyperspectral image classification using semi-supervised learning with label propagation," in *Proc. IEEE India Geosci. Remote Sens. Symp. (InGARSS)*, Dec. 2020, pp. 205–208.

[28] N. Jamshidpour, A. Safari, and S. Homayouni, "A GA-based multi-view, multi-learner active learning framework for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 2, p. 297, Jan. 2020.

[29] Z. Li, H. Huang, Z. Zhang, and Y. Pan, "Manifold learning-based semisupervised neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508712.

[30] Y. Chen, X. Zhao, and X. Jia, "Spectralspatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jan. 2015.

[31] A. Romero, C. Gatta, and G. Camps-Valls, "Unsupervised deep feature extraction for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1349–1362, Mar. 2016.

[32] X. Sun, F. Zhou, J. Dong, F. Gao, Q. Mu, and X. Wang, "Encoding spectral and spatial context information for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2250–2254, Dec. 2017.

[33] X. Zhang, Y. Liang, C. Li, N. Huyan, L. Jiao, and H. Zhou, "Recursive autoencoders-based unsupervised feature learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1928–1932, Nov. 2017.

[34] S. Koda, F. Melgani, and R. Nishii, "Unsupervised spectral–spatial feature extraction with generalized autoencoder for hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 3, pp. 469–473, Mar. 2020.

[35] L. Mou, P. Ghamisi, and X. X. Zhu, "Fully conv-deconv network for unsupervised spectral–spatial feature extraction of hyperspectral imagery via residual learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5181–5184.

[36] Y. Cai, Z. Zhang, Z. Cai, X. Liu, and X. Jiang, "Hypergraph-structured autoencoder for unsupervised and semisupervised classification of hyperspectral image," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[37] Z. Liu, L. Ma, and Q. Du, "Class-wise distribution adaptation for unsupervised classification of hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 508–521, Jan. 2021.

[38] M. Zhang, M. Gong, Y. Mao, J. Li, and Y. Wu, "Unsupervised feature extraction in hyperspectral images based on Wasserstein generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2669–2688, May 2019.

[39] R. Hang, F. Zhou, Q. Liu, and P. Ghamisi, "Classification of hyperspectral images via multitask generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1424–1436, Feb. 2021.

[40] W. Yu, M. Zhang, Z. He, and Y. Shen, "Convolutional two-stream generative adversarial network-based hyperspectral feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506010.

[41] H. Thanh-Tung and T. Tran, "Catastrophic forgetting and mode collapse in GANs," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–10.

[42] H. Xu, W. He, L. Zhang, and H. Zhang, "Unsupervised spectral–spatial semantic feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526714.

[43] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," 2020, *arXiv:2006.11239*.

[44] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. NIPS*, vol. 34, 2021, pp. 8780–8794.

[45] Y. Song and S. Ermon, "Improved techniques for training score-based generative models," in *Proc. Adv. Neural Inf. Process. Syst. 33rd Annu. Conf. Neural Inf. Process. Syst.*, Jan. 2020, pp. 12438–12448.

[46] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674–10685.

[47] N. Liu, S. Li, Y. Du, A. Torralba, and J. B. Tenenbaum, "Compositional visual generation with composable diffusion models," in *Proc. 17th Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2022, pp. 423–439.

[48] Z. Yuan et al., "Efficient and controllable remote sensing fake sample generation based on diffusion model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615012.

[49] Q. Meng, W. Shi, S. Li, and L. Zhang, "PanDiff: A novel pansharpening method based on denoising diffusion probabilistic model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023.

[50] H. Chung, E. S. Lee, and J. C. Ye, "MR image denoising and super-resolution using regularized reverse diffusion," *IEEE Trans. Med. Imag.*, vol. 42, no. 4, pp. 922–934, Apr. 2023.

[51] D. Baranchuk, I. Rubachev, A. Voynov, V. Khrulkov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2021, pp. 1–15.

[52] J. Wu, W. Ji, H. Fu, M. Xu, Y. Jin, and Y. Xu, "MedSegDiff-v2: Diffusion based medical image segmentation with transformer," 2023, *arXiv:2301.11798*.

[53] B. Kim et al., "Diffusion adversarial representation learning for self-supervised vessel segmentation," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–22.

[54] W. G. C. Bandara, N. G. Nair, and V. M. Patel, "DDPM-CD: Remote sensing change detection using denoising diffusion probabilistic models," 2022, *arXiv:2206.11892*.

[55] A. Graikos, N. Malkin, N. Jojić, and D. Samaras, "Diffusion models as plug-and-play priors," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 14715–14728.

[56] E. A. Brempong, S. Kornblith, T. Chen, N. Parmar, M. Minderer, and M. Norouzi, "Denoising pretraining for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 4174–4185.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.

[57] X. Han, H. Zheng, and M. Zhou, "CARD: Classification and regression diffusion models," in *Proc. 36th Conf. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 18100–18115.

[58] Y. Yang, H. Fu, A. I. Avilés-Rivero, C. Schönlieb, and L. Zhu, "DiffMIC: Dual-guidance diffusion network for medical image classification," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Jan. 2023, pp. 95–105.

[59] N. Chen, J. Yue, L. Fang, and S. Xia, "SpectralDiff: A generative framework for hyperspectral image classification with diffusion models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522416.

[60] J. Zhou et al., "Exploring multi-timestep multi-stage diffusion features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519516.

[61] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[62] A. Maffei, J. M. Haut, M. E. Paoletti, J. Plaza, L. Bruzzone, and A. Plaza, "A single model CNN for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2516–2529, Apr. 2020.

[63] J. Zhang, F. Zhao, H. Liu, and J. Yu, "Data and knowledge-driven deep multiview fusion network based on diffusion model for hyperspectral image classification," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123796.

[64] Z. Li et al., "DEMAE: Diffusion-enhanced masked autoencoder for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527616.

[65] Z. Cao, J. Li, and X. Xu, "DiffusionAAE: Enhancing hyperspectral image classification with conditional diffusion model and adversarial autoencoder," *Ecolog. Informat.*, vol. 87, Jul. 2025, Art. no. 103118. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S157495412500127X

[66] S. Prasad and L. M. Bruce, "Limitations of principal components analysis for hyperspectral target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 625–629, Oct. 2008.

[67] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[68] Q. Du and H. Ren, "Real-time constrained linear discriminant analysis to target detection and classification in hyperspectral imagery," *Pattern Recognit.*, vol. 36, no. 1, pp. 1–12, Jan. 2003.

[69] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[70] A. Plaza, P. Martinez, J. Plaza, and R. Perez, "Dimensionality reduction and classification of hyperspectral image data using sequences of extended morphological transformations," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 466–479, Mar. 2005.

[71] F. Tsai and J.-S. Lai, "Feature extraction of hyperspectral image cubes using three-dimensional gray-level cooccurrence," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 6, pp. 3504–3513, Jun. 2013.

[72] L. Sun, Z. Wu, J. Liu, L. Xiao, and Z. Wei, "Supervised spectral–spatial hyperspectral image classification with weighted Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1490–1503, Mar. 2015.

[73] D. Hong, X. Wu, P. Ghamisi, J. Chanussot, N. Yokoya, and X. X. Zhu, "Invariant attribute profiles: A spatial-frequency joint feature extractor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3791–3808, Jun. 2020.

[74] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[75] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[76] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[77] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[78] Z. Zhong, J. Li, L. Ma, H. Jiang, and H. Zhao, "Deep residual networks for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 1824–1827.

[79] X. Tang et al., "Hyperspectral image classification based on 3-D octave convolution with spatialspectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[80] H. Fu et al., "Tensor singular spectrum analysis for 3-D feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403914.

[81] Z. Zhong, Y. Li, L. Ma, J. Li, and W.-S. Zheng, "Spectral–spatial transformer network for hyperspectral image classification: A factorized architecture search framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514715.

[82] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[83] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[84] N. Park and S. Kim, "How do vision transformers work?" in *Proc. Int. Conf. Learn. Represent. ICLR*, 2022, pp. 1–26.

[85] R. Kemker and C. Kanan, "Self-taught feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2693–2705, May 2017.

[86] J. Masci, U. Meier, D. Cireşan, and J. Schmidhuber, "Stacked convolutional auto-encoders for hierarchical feature extraction," in *Artificial Neural Networks and Machine Learning–ICANN 2011*, T. Honkela, W. Duch, M. Girolami, and S. Kaski, Eds., Berlin, Germany: Springer, 2011, pp. 52–59.

[87] B. Du, W. Xiong, J. Wu, L. Zhang, L. Zhang, and D. Tao, "Stacked convolutional denoising auto-encoders for feature representation," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 1017–1027, Apr. 2017.

[88] Y. Zhan, D. Hu, Y. Wang, and X. Yu, "Semisupervised hyperspectral image classification based on generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 212–216, Feb. 2018.

[89] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[90] G. Yang, A. Kundu, L. J. Guibas, J. T. Barron, and B. Poole, "Learning a diffusion prior for NeRFs," 2023, *arXiv:2304.14473*.

[91] M. V. Perera, N. G. Nair, W. G. C. Bandara, and V. M. Patel, "SAR despeckling using a denoising diffusion probabilistic model," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[92] J. Sohl-Dickstein, E. A. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," 2015, *arXiv:1503.03585*.

[93] H. Chung, J. Kim, M. T. McCann, M. Klasky, and J. C. Ye, "Diffusion posterior sampling for general noisy inverse problems," in *Proc. 11th Int. Conf. Learn. Represent.*, Jan. 2022, pp. 1–15.

[94] Z. Cao, S. Cao, X. Wu, J. Hou, R. Ran, and L.-J. Deng, "DDRF: Denoising diffusion model for remote sensing image fusion," 2023, *arXiv:2304.04774*.

[95] X. Li, M. Ding, Y. Gu, and A. Pižurica, "An end-to-end framework for joint denoising and classification of hyperspectral images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 7, pp. 3269–3283, Jul. 2023.

[96] T. Chen, S. Kornblith, M. Norouzi, and G. E. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn.*, Jan. 2020, pp. 1597–1607.

[97] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7482–7491.

[98] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jul. 2015, doi: 10.1155/2015/258619.

[99] Y. Zhu, K. Yuan, W. Zhong, and L. Xu, "Spatial–spectral ConvNeXt for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5453–5463, Jun. 2023.

[100] J. Wang, F. Gao, J. Dong, and Q. Du, "Adaptive DropBlock-enhanced generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5040–5053, Jun. 2021.

[101] L. Huang, Y. Chen, and X. He, "Spectral–spatial masked transformer with supervised and contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508718.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:52:45 UTC from IEEE Xplore. Restrictions apply.